



Comprehensive review and comparative analysis of transformer models in sentiment analysis

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Abstract

Sentiment analysis has become an important task in natural language processing because it is used in many different areas. This paper gives a detailed review of sentiment analysis, including its definition, challenges, and uses. Different approaches to sentiment analysis are discussed, focusing on how they have changed and their limitations. Special attention is given to recent improvements with transformer models and transfer learning. Detailed reviews of well-known transformer models like BERT, RoBERTa, XLNet, ELECTRA, DistilBERT, ALBERT, T5, and GPT are provided, looking at their structures and roles in sentiment analysis. In the experimental section, the performance of these eight transformer models is compared across 22 different datasets. The results show that the T5 model consistently performs the best on multiple datasets, demonstrating its flexibility and ability to generalize. XLNet performs very well in understanding irony and sentiments related to products, while ELECTRA and RoBERTa perform best on certain datasets, showing their strengths in specific areas. BERT and DistilBERT often perform the lowest, indicating that they may struggle with complex sentiment tasks despite being computationally efficient.

Keywords Sentiment analysis · Opinion mining · Transformers · Transfer learning · Text mining · Survey · Comparison study

1 Introduction

The increase in internet use and online activities has created vast amounts of data, known as Big Data. This data is analyzed using data mining, web mining, and text mining techniques for many real-world applications [1]. The large volume of customer opinions and reviews online requires advanced methods to organize and summarize these views [2]. Various online platforms, such as forums, blogs, and social networks, are popular places for expressing opinions, which are important for understanding public sentiment on many topics [3]. In the field of computational linguistics, studying public sentiment, known as sentiment analysis or opinion mining, has become crucial. This area focuses on understanding human emotions,

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attitudes, and opinions related to various topics, such as people, events, products, and services [4]. Sentiment analysis uses methods from machine learning, data mining, and natural language processing, and also gains insights from sociology and psychology [5]. Although natural language processing (NLP) began in the 1950s [6], sentiment analysis gained significant attention from researchers in the mid-2000s [7]. The rise of social media platforms has greatly advanced sentiment analysis. The growth of information on social media platforms has emphasized the importance of sentiment analysis [8]. Its applications are widespread, especially in commerce, where it helps provide recommendations and insights to consumers and businesses. It assists e-commerce platforms in understanding consumer preferences and product perceptions [9]. In politics, sentiment analysis is essential for measuring public opinion and can affect political campaigns and election results [10]. Additionally, it is valuable in public security, as seen in events like the Arab Spring and the London Riots, where social media was crucial [11].

Recently, sentiment analysis has been actively studied by researchers and academics [1]. It involves the computational evaluation of text-based expressions of sentiment toward entities and includes tasks such as detecting, extracting, and classifying sentiments and attitudes [9]. Sentiment analysis helps achieve goals like monitoring public sentiment on political issues, gaining market intelligence, measuring customer satisfaction, and predicting consumer behavior [12]. In e-commerce, customer reviews greatly influence purchasing decisions [13]. Service providers and manufacturers analyze these reviews to improve their products. Studying sentiment analysis in different areas not only improves our understanding of human emotions and opinions but also greatly contributes to achieving various goals, from enhancing consumer satisfaction to shaping public policy. The main contributions of our paper are summarized as follows:

- We performed a detailed comparison of the recent transformer models applied in sentiment analysis, assessing their performance across 22 diverse datasets covering different fields to determine their strengths and weaknesses.
- A thorough overview of sentiment analysis was presented, focusing on defining the problem, outlining challenges, and exploring practical applications in this area. Various methods and approaches utilized for sentiment analysis tasks were discussed. Additionally, the evolution of sentiment analysis techniques over time was traced, and a comparative analysis of these methods was conducted.
- Recent research conducted between 2020 and 2024 was categorized based on its specific applications and the challenges it addresses.
- To enhance reproducibility and support further research in sentiment analysis, we have publicly released the datasets used in this study.¹

The study is structured into five main sections. Section 2 comprehensively discusses sentiment analysis, encompassing definitions, challenges, applications, textual dimensions, and the evolution of methods. Section 3 provides a literature review of various approaches to sentiment analysis, with a focus on transformers and transfer learning models like BERT, RoBERTa, XLNet, ELECTRA, DistilBERT, ALBERT, T5, and GPT. Section 4 presents experimental findings, describing datasets, evaluation metrics, and the performance analysis of these models across a range of datasets. Section 5 addresses limitations and future work, and Sect. 6 concludes the article by summarizing the key findings.

¹ <https://github.com/hadis-1/Sentiment-Analysis-Datasets>.

2 Problem definition, challenges, and applications

Before the main part of this study is discussed, key terms must be defined, and basic background information must be provided. This introductory section will help in understanding Sentiment and Sentiment Analysis, Textual Dimensions in Sentiment Analysis, and the various Applications of Sentiment Analysis. Additionally, it will cover the Evolution of Sentiment Analysis, address Challenges in Binary Sentiment Analysis, and outline the different Levels of Sentiment Analysis.

2.1 Sentiment and sentiment analysis

Emotions and sentiments are key concepts in understanding human behavior [14]. Emotion is a deep psychological state that influences actions and reactions. It includes a range of experiences, from the joy of a surprise birthday party to the grief of loss. These experiences are very personal and can vary widely among individuals [15–17]. Sentiment, however, is more about the judgment or attitude that comes from one's emotions. It is the lasting general feeling or tone toward a particular object or situation [18]. Often influenced by emotions, sentiment is distinct in its persistence and less intense nature [19]. The following hotel review is a hypothetical example to illustrate the difference between emotion and sentiment.

"I recently stayed at the Grandeur Hotel and was thoroughly impressed. From the moment I checked in, the staff was welcoming and attentive. The room was immaculate, with a breathtaking view of the city skyline. The bed was like sleeping on a cloud, and the amenities left nothing to be desired. However, the highlight was the rooftop restaurant, where the culinary delights matched the stunning panoramic views. Despite all these positives, the Wi-Fi connectivity was subpar, which was a slight inconvenience. Overall, the experience was delightful, and the Grandeur Hotel certainly lived up to its name."

In this review, the emotion is the immediate joy and comfort experienced during the stay, while the sentiment is the overall positive impression of the hotel, tempered by a minor negative aspect regarding the Wi-Fi. This sentiment could influence future decisions about staying at the hotel or recommending it to others. Emotions are transient, but the sentiment can have a lasting impact on perceptions and behaviors. Sentiment analysis is the process of determining the emotional tone behind a series of words to understand the attitudes, opinions, and emotions expressed in an online mention. It is also known as opinion mining, deriving the opinion or attitude of a speaker.

Written by: Sara, Date: March 11, 2024: "(1) Recently stayed at the Grandeur Hotel in Paris. (2) Thoroughly impressed with the exceptional service. (3) The room's panoramic view of the Eiffel Tower was breathtaking. (4) The gourmet dining options exceeded my culinary expectations. (5) However, according to my mother, who is quite tech-savvy, the in-room technology was somewhat outdated and not user-friendly."

Analyzing this review, the following components can be identified in Table 1:

2.2 Textual dimensions in sentiment analysis

Sentiment analysis, a key aspect of NLP, examines human emotions expressed through text. Interpreting sentiments computationally requires understanding the textual elements that represent these emotions. An in-depth look at these elements will be made, specifically

Table 1 Analyzing a review

Components	Values
Source of opinion	Sara for statements (1)–(4), and Sara’s mother for statement (5)
Date of review	March 11, 2024
Subject	Grandeur Hotel in Paris
Features	Hotel service, room view, dining options, technology
Sentiment	A mix of positive and negative impressions of the Grandeur Hotel. Positive remarks are apparent in statements (2)–(4), highlighting the service and dining as exceptional, while a critical view of the hotel’s technology, as shared by Sara’s mother, is expressed in statement (5)

focusing on subjectivity vs. objectivity, explicit vs. implicit expressions, and simple vs. comparative opinions.

1. **Subjectivity versus Objectivity opinions:** The distinction between subjective and objective texts is essential in sentiment analysis. Subjective texts are filled with personal stories, emotional expressions, and evaluative language, directly showing the author’s sentiments. These texts are crucial for sentiment analysis, providing rich, qualitative data for insights into public opinion, consumer preferences, and cultural trends. On the other hand, objective texts are neutral and fact-based, often presenting statistical data, historical information, or technical details. Although not overtly emotional, they can carry subtle sentiments influencing public perception and decision-making [20, 21].
2. **Explicit versus Implicit opinions:** Explicit and implicit texts differ in how directly sentiments are expressed. Explicit texts clearly show their sentiments, making them easy for sentiment analysis algorithms to classify and measure. Implicit texts, however, are more complex and require careful analysis of context, word choice, and subtext to uncover the underlying sentiments [21].
3. **Simple versus Comparative opinions:** Simple opinions are the basic units of sentiment analysis, offering straightforward evaluations of entities or their attributes. They facilitate easy categorization and trend analysis. Comparative opinions, however, add complexity by evaluating entities or attributes in relation to one another. This requires analyzing both the sentiments and the comparative structures that define the relationships between entities or attributes [22].

2.3 Sentiment analysis applications

The number of publications with terms like “sentiment analysis,” “sentiment classification,” “opinion mining,” and “polarity detection” in their titles or abstracts has been extracted, as shown in Fig. 1. As its various uses are explored, its important role in enhancing the intelligence of businesses and organizations is recognized. This section covers the practical applications of sentiment analysis in different industries, highlighting its effect on improving business operations and decision-making. The four main categories of sentiment analysis applications, as identified by research in this field, are examined.

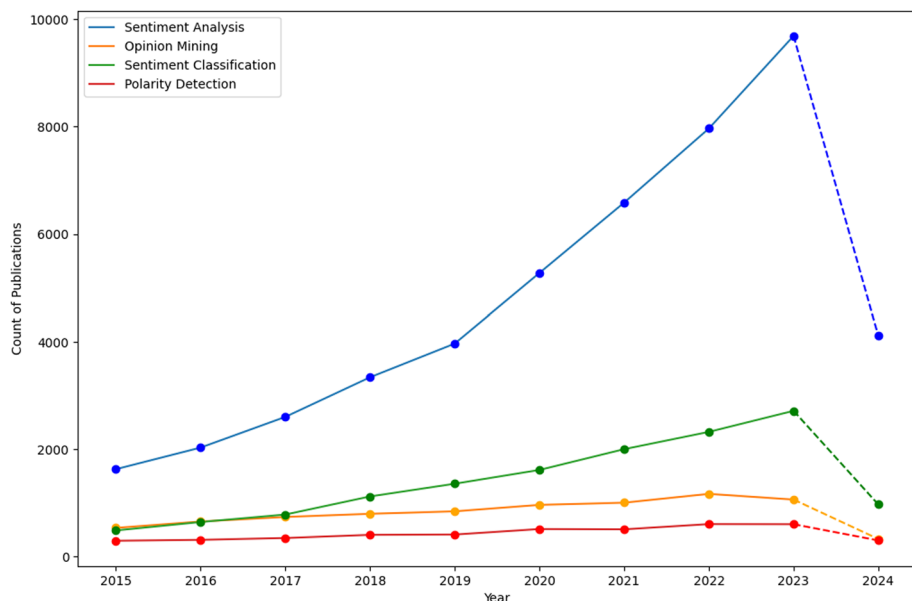


Fig. 1 Count of publications that include specific keywords in their titles or abstracts

2.3.1 Customer service

Sentiment analysis is now a key tool in customer service for understanding and sorting customer sentiments from text feedback. This technology helps service teams to prioritize issues and customize responses, improving customer satisfaction and guiding business strategies. Here is a detailed look at its applications [23, 24]:

- **Understanding customer sentiments:** The main role of sentiment analysis in customer service is to grasp the emotions of customers as they interact with a company. Whether through social media, reviews, support tickets, or direct feedback, these tools can detect if sentiments are positive, negative, or neutral. This information is vital for customer service teams as it helps them prioritize which issues to address first, especially with a high volume of interactions.
- **Enhancing response strategies:** By identifying the sentiment behind customer inquiries, representatives can adjust their responses to match the customer's emotional state. For example, a frustrated customer due to a delayed shipment needs a different approach than one simply seeking information. Sentiment analysis helps craft responses that are both informative and empathetic, greatly improving the customer experience.
- **Measuring customer satisfaction:** Sentiment analysis is also used to measure overall customer satisfaction over time. By tracking the sentiment of interactions, companies can spot trends and patterns indicating broader issues with products or services. This can lead to proactive changes in business strategies to enhance satisfaction and reduce churn.
- **Training and quality assurance:** Sentiment analysis is also important for training and quality assurance of customer service teams. By examining the sentiment of interactions, managers can find areas where representatives need more training. It also acts as a quality control tool, ensuring all interactions meet company standards.

- **Real-time feedback for product and service development:** Sentiment analysis offers real-time feedback for product and service development. Understanding how customers feel about features or services helps companies make improvements or create new offerings that better meet needs.
- **Product recommendations:** Sentiment analysis can suggest products or services that align with the customer's sentiment and preferences. By analyzing sentiment and context, businesses can find opportunities to recommend products that may interest the customer, increasing potential sales.

2.3.2 Finance and stock market

In finance, sentiment analysis is used to guide investors by understanding market opinions. It measures the general mood of market participants, giving an overview of attitudes towards financial instruments. This emotional measure can be especially useful in the stock market, where investor sentiment often predicts market movements. In finance and stock market analysis, sentiment analysis is not just a tool; it is a key asset. It uses vast amounts of unstructured data—news articles, social media, financial blogs, and analyst reports—to uncover subtle signals driving market trends. Here's how it works [25]:

- **Market prediction:** Sentiment analysis can predict market direction. Positive sentiment may suggest a rise in stock prices, while negative sentiment could indicate a drop. By examining the tone of financial news and expert opinions, investors can foresee market movements before they happen. For example, a rise in positive sentiment towards a particular industry might suggest upcoming growth, prompting investors to buy stocks in that sector before prices increase.
- **Regulatory compliance:** With financial regulations in place, sentiment analysis helps compliance officers. By monitoring sentiment, unusual patterns indicating market manipulation or insider trading can be detected, ensuring market integrity. For instance, a sudden, unexplained rise in positive sentiment and trading volume in a small stock might signal possible manipulation, leading to further investigation to maintain fair trading practices.
- **Investor relations:** Companies monitor sentiment to understand investor perceptions and expectations. This feedback can guide corporate strategies, communication plans, and decisions on mergers and acquisitions. For example, if sentiment analysis reveals investor concerns about a company's environmental impact, the company might prioritize sustainability initiatives and communicate these efforts to improve public perception and investor confidence.
- **Risk management:** Understanding sentiment helps manage risk. A sudden shift in sentiment can signal potential volatility. Investors tracking these changes can adjust their portfolios to minimize potential losses. For instance, if sentiment analysis shows increasing negativity around a company due to scandals or poor financial reports, investors might reduce their holdings in that company to avoid substantial losses if the stock price drops.

2.3.3 Politics

In politics, sentiment analysis is a tool, turning public opinion into actionable insights. It helps campaign strategists measure voter sentiment by analyzing tweets, posts, and comments. Here's how it is applied [26, 27]:

- **Predicting election outcomes:** By analyzing data from news articles, social media, and public forums, sentiment analysis gauges public opinions about political candidates or parties. This can forecast electoral trends, providing campaigns with insights into voter behavior and preferences. As a result, sentiment analysis is essential in modern political campaigns.
- **Voter sentiment tracking:** This involves analyzing social media data to provide real-time feedback on public opinion. Campaigns can track changes in voter sentiment towards policies, candidates, or political events, allowing proactive responses. For example, if negative sentiment rises after a policy announcement, the campaign can quickly address concerns and adjust messaging to align with voter expectations.
- **Issue identification:** Sentiment analysis helps campaigns identify which issues resonate most with voters. Politicians can then prioritize policy positions and tailor messages to match voter sentiment. For instance, if climate change is a top issue among voters, the campaign can focus on environmental policies and highlight their candidate's plans to address climate issues.
- **Opposition research:** By monitoring public perception of opponents, campaigns can find potential weaknesses or areas of concern. This information can be used to strengthen their position or counter opponents' strategies. For example, if an opponent faces backlash over a controversial statement, the campaign can highlight their candidate's differing stance to gain favor with undecided voters.
- **Get-Out-The-Vote efforts:** Understanding voter sentiment informs get-out-the-vote strategies. Campaigns can identify areas of strong support or apathy and focus efforts where needed most to maximize voter turnout. For instance, if certain demographics show high support but low voter registration, targeted efforts can encourage registration and participation in the election.
- **Crisis management:** In political controversies, sentiment analysis assesses public reaction. This enables rapid response and damage control to maintain a candidate's public image and reduce negative impacts. For example, if a candidate is criticized for a policy decision, the campaign can quickly issue statements, adjust their stance, and engage with the public to manage the narrative and mitigate negative fallout.

2.3.4 Healthcare and medical

In healthcare, sentiment analysis acts as a diagnostic tool, examining patient feedback to find meaningful patterns. It goes beyond traditional metrics of care quality, exploring patients' emotional responses for a comprehensive view of healthcare experiences. Sentiment analysis in healthcare and medical reviews uses patient sentiments to improve care delivery. Here's how it is used [28–30]:

- **Patient satisfaction:** Sentiment analysis examines patient reviews to find insights on satisfaction levels. It considers both quantitative ratings and qualitative emotions and experiences. This detailed understanding helps healthcare providers identify specific aspects of care that please or displease patients. For instance, analysis might show that patients appreciate staff friendliness but are unhappy with long waiting times. Addressing such concerns can significantly improve overall patient satisfaction.
- **Treatment efficacy:** Sentiment analysis captures patients' experiences with treatments and medications, revealing how they feel about treatment effectiveness and side effects. For example, patients might be frustrated with side effects despite a treatment being clinically effective, prompting healthcare providers to consider alternatives or additional support.

Table 2 Categorization of conducted research for various sentiment analysis applications from 2020 to 2024

Sentiment analysis applications	Researches
Customer Service	[12, 31–91]
Finance and stock market	[92–128]
Politics	[10, 129–155]
Healthcare and medical	[28, 156, 157, 157–187]

- **Service improvement:** Healthcare services continuously evolve, and sentiment analysis helps drive this change. By highlighting what patients appreciate and dislike, healthcare facilities can innovate and adapt to provide exceptional services. For example, positive feedback on telemedicine services might encourage expanding these offerings, while negative feedback on appointment scheduling might lead to process improvements.
- **Policy formulation:** In healthcare policy-making, sentiment analysis guides the creation of policies that resonate with the public. It ensures reforms are informed by the voices of those affected. For instance, if sentiment analysis shows concern about mental health service accessibility, policymakers might prioritize improving mental health care infrastructure and funding.
- **Clinical decision support:** In clinical decision-making, sentiment analysis adds emotional intelligence, ensuring decisions are patient-centered. For example, if feedback highlights concerns about treatment side effects, clinicians can consider this along with clinical data for more informed decisions.
- **Quality control:** Sentiment analysis continuously measures patient feedback, providing ongoing insights into care quality. It's a tool for constant improvement, ensuring care standards rise. For instance, real-time feedback analysis can help a hospital quickly identify and address issues, such as increased negative feedback about a particular department or service.

In addition to these applications, sentiment analysis is valuable in various sectors. In entertainment, it helps understand audience responses. In human resources, it assesses employee satisfaction and improves recruitment. It gauges public opinion for policy formulation and societal impact assessment. In education, it informs curriculum development and improves student feedback analysis. The travel industry uses sentiment analysis to enhance customer experiences. For new research focus, Table 2 categorizes studies related to each application of sentiment analysis from 2020 to 2024.

2.4 The evolution of sentiment analysis

The field of sentiment analysis has advanced from simple binary classification to methods that reflect the complexities of human emotions. This evolution has resulted in the creation of various specialized approaches to capture subtle sentiment nuances in different contexts [30]. Here, we will explore the distinct categories of sentiment analysis, starting with basic binary classification and moving to more complex multimodal sentiment analysis.

2.4.1 Binary sentiment analysis

Binary sentiment analysis is the basic approach in sentiment analysis. It is ideal for applications where the main goal is to identify a clear positive or negative sentiment, such as

monitoring brand sentiment on social media or analyzing customer feedback. Its simplicity allows for high scalability and efficiency in real-time sentiment tracking. Businesses can use it for real-time monitoring of social media to gauge public reactions to marketing campaigns or new product launches. Customer feedback systems can quickly sort reviews into positive and negative categories, enabling swift responses to customer concerns and helping maintain brand reputation [188–190].

2.4.2 Fine-grained sentiment analysis

Fine-grained sentiment analysis is useful when the intensity of the sentiment needs to be captured. For example, understanding whether customers are mildly pleased or extremely satisfied can influence product development and marketing. This method is also valuable in customer service to prioritize responses based on the intensity of sentiment in customer communications. This type of analysis helps businesses understand not just the sentiment but its strength, enabling more nuanced and effective decision-making. For instance, a very dissatisfied customer may need immediate attention compared to one who is only mildly displeased. In product development, knowing if feedback is strongly or weakly positive can guide priorities in feature enhancements or new product launches [188–190].

2.4.3 Emotion recognition

Emotion recognition goes beyond basic sentiment analysis to identify specific emotions such as joy, anger, sadness, and surprise. This type of analysis is especially useful in fields like mental health monitoring, where understanding an individual's emotional state can provide critical insights. It is also used in interactive applications like virtual assistants or customer service bots to recognize user emotions and enhance interactions. Emotion recognition is important in various domains. In mental health, identifying emotions in patient communications can help in early diagnosis and timely intervention. In virtual assistants, detecting user emotions like frustration or joy allows for more personalized and engaging interactions, improving user satisfaction and loyalty [191].

2.4.4 Multimodal sentiment analysis

Multimodal sentiment analysis is advanced and is best used in environments with multiple forms of communication, such as video reviews, vlogs, or customer support calls. It provides a comprehensive view of sentiment by analyzing verbal cues along with non-verbal ones like facial expressions and tone of voice. Analyzing both verbal and non-verbal cues allows for a deeper understanding of sentiment. In customer support, recognizing a frustrated tone in a customer's voice in addition to their words can prompt more empathetic and effective responses. In content analysis, combining verbal content with facial expressions can provide deeper insights into the creator's sentiments [192, 193].

This research focuses on binary sentiment analysis due to its scalability and efficiency, making it essential for real-time applications.

2.5 Challenges in binary sentiment analysis

Binary sentiment analysis faces several challenges that affect the accuracy of sentiment categorization. These challenges include linguistic subtleties, contextual dependencies, dynamic

language evolution, and cultural divergences. Below are detailed descriptions of these challenges.

2.5.1 Linguistic quandaries

Binary sentiment analysis faces significant challenges due to the ambiguity and nuances of natural language. Sentiments are subtle and depend on context, making it hard for algorithms to accurately classify text. Sarcasm, irony, and idiomatic expressions are especially problematic, as they often express sentiments opposite to their literal meanings. For example, “Great, just what I needed!” can be positive or highly negative depending on the context. If someone gets an unexpected work assignment, they might sarcastically say, “Great, just what I needed!” The words suggest positivity, but the tone and context indicate annoyance or burden. Similarly, idiomatic expressions, which are deeply rooted in cultural and linguistic contexts, can confuse sentiment analysis algorithms. These expressions carry meanings beyond their literal words, complicating sentiment interpretation. For example, the idiom “raining cats and dogs” might confuse systems unfamiliar with English idioms. While native speakers understand it means heavy rainfall, a literal interpretation could mislead an algorithm, causing it to misclassify the sentiment. Handling sarcasm, irony, and idioms requires more than simple lexical analysis [191–193].

Polysemy, where one word has multiple meanings, adds further complexity to sentiment classification. This phenomenon challenges models by presenting different interpretations of a word based on context. For example, “cool” can mean temperature, social approval, or emotional detachment depending on the situation. The ambiguity of polysemous words like “cool” highlights the need for advanced disambiguation mechanisms in sentiment analysis. Without these mechanisms, models may misinterpret such terms, leading to incorrect sentiment classification [194].

2.5.2 Domain-specific

Sentiment expression varies significantly across different domains, adding more complexity. For instance, the language in financial news differs greatly from that in social media or product reviews. Domain-specific jargon, formality levels, and contextual cues all affect sentiment interpretation. Models trained on one type of data often perform poorly on another, showing the need for domain adaptation techniques and robust, diverse training datasets. Consider the difference between the emotive language in financial reports and the language used in social media discussions or product reviews. Each domain has specialized terminology, different levels of formality, and unique contextual indicators that shape sentiment interpretation. Consequently, algorithms tailored to one dataset often fail with different data, highlighting the need for domain adaptation and diverse training data [195, 196].

2.5.3 Data availability

A lack of labeled data across domains poses another challenge. This scarcity hampers model performance due to inadequate training data for each domain. For example, in healthcare, there may be abundant labeled data for customer reviews or social media, but not enough for sentiments in medical forums or patient feedback. This lack of domain-specific labeled data makes it hard to train accurate models that capture nuances across various domains [8, 197].

2.5.4 Dynamic evolution of linguistic usage

Language constantly evolves, with new slang, expressions, and modes of communication like emojis and hashtags emerging. This continuous change presents a challenge for maintaining effective sentiment analysis models. Static models that don’t adapt to linguistic shifts quickly become obsolete, making regular retraining with updated datasets essential for maintaining accuracy and relevance. For example, internet slang evolves over time, with phrases like “LOL” (laugh out loud) or “ICYMI” (in case you missed it) becoming common. These phrases convey nuanced sentiments that traditional algorithms may not capture. Additionally, emojis add a visual dimension to text, allowing users to express emotions more subtly. Ignoring these evolving linguistic conventions can reduce model accuracy and relevance [198]. Table 3 shows the categorization of research conducted from 2020 to 2024 for specific sentiment analysis challenges.

2.6 Sentiment analysis levels

Sentiment analysis has been studied at several levels: Document Level, Sentence Level, Phrase Level, and Aspect Level. Each level has its own applications [9, 198, 284]. These levels are shown in Fig. 2.

Table 3 Categorization of conducted research for various sentiment analysis challenges from 2020 to 2024

SA challenges	Researches
Linguistic quandaries	[199–201, 201–214]
Domain-specific	[8, 195, 215–233]
Data availability	[234–239, 239–262]
Dynamic evolution of linguistic usage	[263–283]

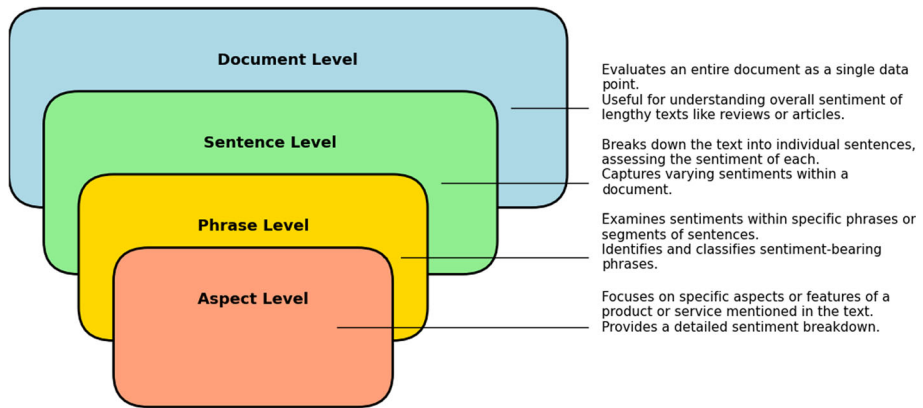


Fig. 2 Levels of sentiment analysis

3 Literature review

This section begins by explaining various methodologies used in sentiment analysis, examining research efforts within each approach. Then, the limitations of traditional machine learning and deep learning models are discussed, explaining why transformers are widely adopted in sentiment analysis. The structural composition of transformers is also explained, highlighting the advantages of using these advanced models. Additionally, a thorough review of prominent transformer architectures is conducted in the results section.

3.1 Sentiment analysis approaches

Sentiment analysis includes various methodologies, each with its approach. These methodologies include lexicon-based methods, machine learning, deep learning techniques, hybrid methods, and transfer learning. These categories are shown in Fig. 3. Each method has unique advantages and challenges, which will be analyzed in the following sections.

3.1.1 Lexicon-based approach

Lexicon-based sentiment analysis involves assigning predefined scores to words indicating their polarity (positive, neutral, or negative). These scores are aggregated to determine the overall sentiment of a text. While this method doesn't require training data, it's highly dependent on domain, as word polarity can vary across contexts [4]. For example, "small" can be negative when referring to a TV screen size but positive when describing a camera's portability. Domain-specific lexicons or adaptations can mitigate this issue. The approach has advantages such as being unsupervised but is limited by its domain specificity. Two main approaches exist: Corpus Based and Dictionary based method [285], each with its own advantages and disadvantages.

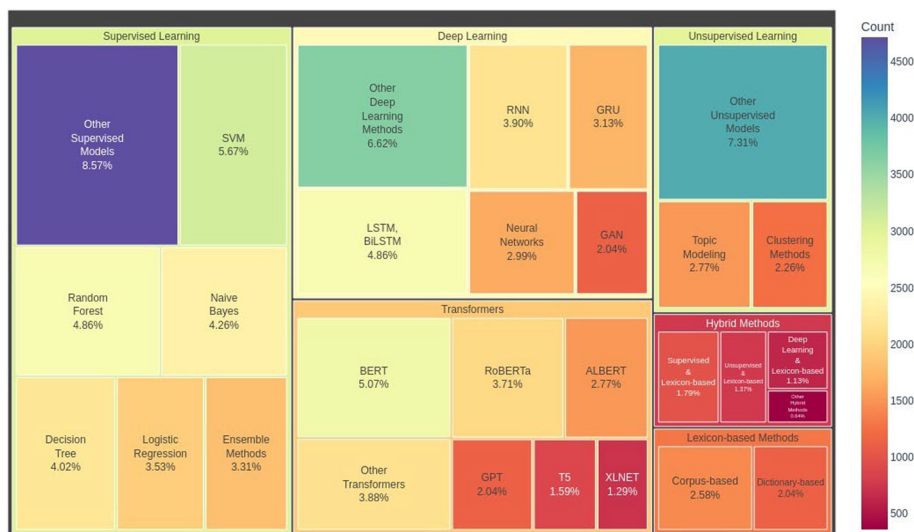


Fig. 3 Sentiment analysis approaches

The corpus-based approach in sentiment analysis utilizes semantic and syntactic patterns to determine the sentiment of sentences. It relies on predefined sentiment terms and their orientation, analyzing large corpora to identify sentiment tokens and their orientations. While this method is context-specific and requires labeled data for training, it addresses the challenge of opinion words with context-dependent orientations. For example, [286] employed linguistic constraints and connectives to infer sentiment from tokens based on surrounding words. They constructed a graph representation of tokens and their relationships, using a linear log model to determine sentiment orientations. The corpus-based approach encompasses Statistical and Semantic approaches, with Statistical methods utilizing co-occurrence patterns to infer sentiment orientations. [287] proposed calculating mutual information to assess the frequency of token co-occurrences, often applied in detecting manipulated reviews. For instance, [288] utilized statistical tests to identify manipulated customer reviews on Amazon. Additionally, Latent Semantic Analysis (LSA) has been used to uncover semantic patterns in reviews. [289] utilized LSA to analyze user feedback from CNETdownload.com, investigating factors influencing review helpfulness ratings. The Semantic Approach in sentiment analysis involves calculating similarity scores between tokens, often using resources like WordNet. This method enables the identification of antonyms and synonyms, as similar words typically have positive similarity scores. [290] proposed utilizing this approach to construct lexicon models for sentiment analysis, providing detailed descriptions of subjectivity relations among characters in textual statements. They emphasized the importance of incorporating knowledge about the identity and attitude orientation of individuals expressing opinions. Researchers such as [291, 292], and [292] have utilized WordNet datasets in their work to distinguish between the subjectivity of viewers and actors in certain contexts. This approach offers a nuanced understanding of sentiment by considering the complex relationships between words and their meanings.

The dictionary-based approach in sentiment analysis relies on predefined sets of opinion words, manually collected and categorized with their respective polarities. This method assumes that synonyms share the same polarity as the base word, while antonyms have opposite polarities. Initial word sets are expanded by incorporating antonyms and synonyms from lexical resources such as thesauri or WordNet. Manual evaluation and correction are often employed to ensure list quality. SentiWordNet, developed by [293], is an example of such a resource, created through automatic annotations of WordNet 3 synsets. [286] proposed a rule-based method for sentiment labeling in contextual advertising using a dictionary-based approach, suitable for small dictionaries. However, a challenge in all lexicon-based approaches, including the dictionary-based approach, is the need for domain-specific opinion words due to varying polarities across different contexts. Table 4 provides a comparative analysis, highlighting the advantages and disadvantages of these two categories of methods.

3.1.2 Machine learning and deep learning techniques

In recent years, machine learning (ML) and deep learning (DL) have emerged as pivotal domains within the realm of artificial intelligence and data science. ML algorithms empower computers to learn from and make predictions or decisions based on data, while DL techniques, inspired by the structure and function of the human brain, delve deeper into complex data representations through hierarchical layers of abstraction. In this section, we introduce the research conducted to analyze emotions using these approaches. [294] were among the first to use machine learning techniques such as NB, Maximum Entropy, and SVM for sentiment analysis of movie reviews. They gathered reviews from IMDb.com and experimented with different feature engineering methods. SVM achieved the highest accuracy of 82.9%

Table 4 Advantages and disadvantages of dictionary-based and corpus-based approaches

	Dictionary-based approach	Corpus-based approach
Advantages	No need for extensive training data	Considers contextual nuances and syntactic patterns
	Provides straightforward polarity assignment for words	Incorporates semantic relationships between tokens
	Can be implemented relatively quickly and efficiently, especially for smaller dictionaries	Offers a nuanced understanding of sentiment through semantic analysis
Disadvantages	Limited coverage and scalability due to dependency on manually curated word lists	Requires significant labeled data for training, making it less feasible for certain applications
	Domain-specific adaptation is challenging, as word polarity may vary across contexts	Time-consuming and computationally intensive, especially for large corpora
	Prone to inaccuracies and biases inherent in the chosen dictionary	May struggle with ambiguous language and evolving linguistic trends

using unigrams. They suggested that incorporating discourse analysis, focus detection, and co-reference resolution could further enhance accuracy. [295] introduced a structured model for joint sentiment classification at sentence and document levels. They represented documents and sentences as cliques and utilized the Viterbi algorithm for inference. MIRA algorithm was used for classification, considering features like unigrams, bigrams, trigrams, and POS. They tested their model on 600 product reviews from Amazon.com across three domains, achieving accuracies of 62.6% at the sentence level and 82.8% at the document level. [296] also employed SVM for sentiment classification, conducting experiments on digital camera reviews and a multi-domain dataset. They trained SVM on different feature sets and applied Information Gain (IG) for feature selection, achieving an accuracy of 84.15% for kitchen appliances. [297] conducted extensive sentiment classification experiments using SVM and various feature selection methods on different datasets, reporting accuracies ranging from 73.25 to 91.51%. [298] proposed a Tabu-Search-Markov-Blanket Classifier (TS-MBC) which achieved accuracies of 78.08%, 92.70%, and 73.21% on different datasets. [299] focused on sentiment classification of Cantonese restaurant reviews using NB and SVM, achieving high accuracy with different feature representations. [300] introduced an automatic approach for sentiment polarity detection at the phrase level, achieving high F-measure. [301] compared ensemble methods for sentiment classification and reported improved accuracy over base learners. [302] compared SVM and NB with ANN-based approach on various datasets, observing the impact of feature selection on performance. [6] Proposed a hybrid PSO-SVM method for sentiment classification, achieving promising results. [303] developed a Twitter-specific lexicon for sentiment analysis using supervised feature reduction and statistical analysis, achieving high accuracy with DAN2. They suggested incorporating an ontological approach for further improvement.

[304] proposed a novel neural network architecture for learning vector-based document representations. This model adopts a unified, bottom-up approach where sentence representations are learned using either convolutional neural networks or long short-term memory. The

semantics of sentences and their relations are then encoded adaptively in the document representation using gated recurrent neural networks. In a separate work, [304] introduced a deep learning system called Coooolll for message-level Twitter sentiment classification. Coooolll integrates sentiment-specific word embedding (SSWE) features with traditional hand-crafted features in a supervised learning framework. SSWE is trained from large-scale tweet data without manual annotation, making it easily re-implementable. [304] addressed the limitation of existing word embedding algorithms, which typically only capture syntactic context, by presenting a method for learning SSWE. These embeddings encode sentiment information within the continuous representation of words. Additionally, they developed three neural networks to incorporate supervision from sentiment polarity in the loss functions effectively. [305] introduced a cascaded framework of feature selection and classifier ensemble using particle swarm optimization (PSO) for aspect-based sentiment analysis. This framework performs aspect term extraction and sentiment classification in two steps. The compact set of features obtained through feature selection outperformed the baseline model using the complete feature set. [306] investigated the effect of using regression on confidence scores in sentiment analysis using Turkish tweets. They extracted hand-crafted features, including lexical features and emoticons, and employed word embeddings for regression and classification. The findings suggest that employing regression on confidence scores can slightly improve sentiment classification accuracy. Additionally, combining word embeddings with hand-crafted features can reduce feature dimensionality and outperform alternative feature combinations.

3.1.3 Hybrid methods

Researchers have explored various approaches to sentiment analysis across different domains. Sindhwani et al. [306] developed unsupervised and semi-supervised sentiment classification algorithms. They utilized a lexicon for domain adaptability and employed Regularized Least Squares (RLS) classification due to data sparseness. Their methods showed promising results across diverse domains. [306] conducted linguistic analysis and sentiment classification using supervised learning. They utilized various linguistic features to determine sentence orientation and achieved satisfactory results, although further experimentation is warranted. [307] developed hybrid classifiers and achieved high accuracy by combining different methods. Their approach requires validation on larger datasets for robustness. [308] introduced a Feature Relation Network (FRN) for sentiment classification, outperforming other techniques on multiple datasets. Future research avenues include augmenting FRN with additional features. [309] explored ensemble techniques for sentiment classification, achieving high accuracy by combining different feature sets with machine learning algorithms. Feature selection based on syntactic relations could further enhance performance. [310] proposed a hybrid approach for sentiment classification of product blogs, showing promising results that warrant testing on diverse datasets and platforms. [311] extended her method for emotion detection using commonsense knowledge, outperforming other techniques. Future work includes extending EmotiNet to other languages and domains. [312] conducted subjectivity and sentiment analysis of social media in a morphologically-rich language, achieving high accuracy with SVM. Error analysis and irony detection could further improve accuracy. [313] developed a sentiment lexicon and employed machine learning algorithms for sentiment classification on Facebook comments, showing promising results. Future improvements could include calculating contextual sentiment scores. [314] proposed a target-dependent sentiment classification approach for tweets, achieving high accuracy

through graph-based optimization. Future research could explore user profile-based sentiment classification. [314] employed emotion cause extraction and chi-square method for emotion classification in tweets, achieving satisfactory precision. Future work could involve automatic generation and modification of pattern sets to reduce manual effort. [315] proposed early detection of Twitter trends, showing promising results for detecting emerging topics during events like elections. Future research could explore emerging patterns of controversial topics. Table 5 provides a detailed categorization and description of the advantages and disadvantages of the mentioned approaches.

3.2 Transformers and transfer learning

Sentiment analysis started with simple methods like Naive Bayes or SVM algorithms [316]. As technology progressed, artificial neural networks, especially architectures like LSTM and CNN, were used. These methods, though somewhat effective, often struggled to capture long-range dependencies and contextual nuances, requiring extensive feature engineering. The introduction of transformers solved these issues, providing a robust solution and enhancing the ability to understand texts comprehensively. Recently, researchers have increasingly used transformer models, revolutionizing sentiment analysis with innovative self-attention mechanisms and bidirectional processing capabilities [317]. This section explores the reasons for this shift and examines the advantages of transformer models in sentiment analysis.

3.2.1 Limitations of traditional approaches

Previous sentiment analysis methods faced significant limitations due to reliance on manually engineered features in conventional machine learning algorithms. These features often failed to capture the intricate nuances of language, limiting the models' ability to accurately discern sentiment [14]. Early deep learning architectures like RNNs and CNNs also struggled to effectively capture long-term dependencies and contextual information in textual data, resulting in suboptimal performance in sentiment analysis tasks [318]. Traditional machine learning algorithms typically required domain-specific feature engineering, where experts manually crafted features relevant to sentiment classification. However, this approach was inherently limited by human bias and often failed to capture the full complexity of language, reducing accuracy and generalizability. Similarly, early deep learning models like RNNs and CNNs, although promising in learning representations directly from raw text data, faced challenges in effectively modeling sequential dependencies and contextual information over long input sequences. RNNs suffered from vanishing gradient problems, making it difficult to capture long-range dependencies effectively [319], while CNNs excelled in capturing local patterns but struggled with global context in text data [320]. These limitations prompted researchers to explore more advanced techniques for sentiment analysis, leading to the development of transformer architectures. The introduction of transformer models marked a significant shift in sentiment analysis methodologies [321]. Transformers, introduced in the seminal paper "Attention is All You Need" by Vaswani et al. [322] in 2017, brought a new approach to handling sequential data. The key innovation of transformers is the self-attention mechanism, which allows the model to weigh the importance of different words in a sentence, effectively capturing contextual relationships over long distances. This mechanism addresses the shortcomings of previous models and enables more accurate sentiment analysis.

Table 5 Comparison of SA approaches with their advantages and disadvantages

	Advantages	Disadvantages
Machine learning approaches	Can achieve high accuracy when feature engineering is done well, enabling precise sentiment classification	Requires a substantial amount of labeled data for training, which can be costly and time-consuming to obtain
	Effective for handling large datasets with diverse features, allowing for scalability and robustness in analysis	Performance is highly sensitive to the choice of features and the tuning of algorithm parameters, which can impact results if not optimized correctly
	Supports a variety of algorithms such as SVM, NB, and Maximum Entropy, offering flexibility in method selection based on specific needs	The effectiveness of these algorithms can vary significantly depending on the domain and dataset, leading to inconsistent performance across different scenarios
	Can be somewhat robust to noisy data, as many algorithms include mechanisms to handle or reduce the impact of noise during analysis	May not capture deep semantic nuances or context without extensive feature engineering, limiting their ability to understand complex or subtle sentiments
Deep Learning approaches	Capable of learning and representing complex patterns and relationships within data, which enhances the accuracy of sentiment predictions	Requires very large labeled datasets for effective training, making it difficult to implement without substantial data resources
	Often achieves high accuracy, particularly with large datasets, due to the advanced capabilities of deep learning models in capturing intricate patterns	Computationally expensive and time-consuming to train, requiring powerful hardware and significant time investment
	Automatically extracts and learns features from raw data, reducing the need for manual feature selection and engineering, which simplifies the analysis process	Models can overfit on small datasets or if regularization is not properly applied, potentially reducing their generalizability to new data
	Utilizes models like LSTM and CNN that effectively capture context and semantics, improving sentiment analysis results	Deep learning models are often less interpretable compared to traditional machine learning methods, making it harder to understand and explain model decisions
	Capable of learning and representing complex patterns and relationships within data, which enhances the accuracy of sentiment predictions	Requires very large labeled datasets for effective training, making it difficult to implement without substantial data resources

Table 5 (continued)

	Advantages	Disadvantages
Hybrid approaches	Combines the strengths of multiple methods, leading to enhanced performance and improved accuracy by leveraging the benefits of different techniques	Implementing hybrid approaches can be complex, as it involves integrating different methods and managing their interactions, which can be challenging
	Handles various types of data and features more effectively by incorporating a diverse set of techniques, improving overall analysis capabilities	Requires extensive validation and tuning to ensure optimal performance, which can be resource-intensive and time-consuming
	Often results in higher robustness and accuracy compared to single-method approaches, as it benefits from the complementary strengths of different methods	Computationally intensive, particularly when dealing with large datasets or complex models, necessitating significant computational resources and expertise
	Flexible in integrating domain-specific knowledge with general models, making it adaptable to different contexts and improving analysis results	The complexity of managing and optimizing multiple methods can be a barrier, requiring careful planning and expert knowledge to achieve the best results

3.2.2 Transformers in SA task

The transformer, a groundbreaking neural architecture developed by Google researchers in 2017, has become essential in various domains, including NLP, machine translation, text generation, and named entity recognition. Using self-attention mechanisms, transformers capture intricate word relationships within sentences, enabling efficient parallel processing of data sequences [321]. A key feature of transformers is the multi-head self-attention mechanism, which allows simultaneous focus on different segments of input sequences, capturing both local nuances and global dependencies [323]. This parallelization and attention mechanism make transformers exceptionally proficient in processing sequential data. The use of transformers in sentiment analysis offers numerous advantages [324]. These models excel in identifying relationships within lengthy sentences, crucial for nuanced sentiment interpretation across extended texts. Additionally, transformers enable parallel processing, significantly speeding up both training and inference, which is particularly beneficial for managing large datasets [325]. Their attention mechanisms prioritize relevant words, enhancing sentiment analysis accuracy by focusing on pivotal sentiment-bearing phrases. A notable advancement in transformer models is their adaptability through fine-tuning pre-trained models for specific tasks. Models like BERT and GPT undergo initial training on extensive datasets followed by fine-tuning with task-specific data, leveraging the knowledge encoded during pre-training to achieve precision with minimal labeled data. Transfer learning and domain adaptation are crucial strategies for leveraging transformer models in sentiment analysis. Transfer learning allows models to apply knowledge across domains, minimizing the need for extensive labeled data in target domains. This is particularly advantageous in sentiment analysis, enabling models to adapt to diverse industries or subjects with minimal additional training [326]. As transformer models evolve, the demand for explainability and interpretability grows.

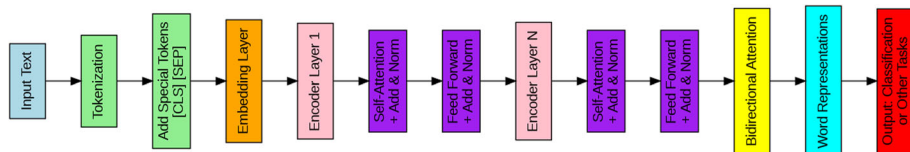


Fig. 4 Architecture of the BERT model

Researchers are developing techniques to clarify the inner workings of these models, providing insights into their decision-making processes. Methods like SHAP and LIME enhance interpretability [327], increasing the transparency of transformer-based sentiment analysis models.

3.2.3 Bidirectional encoder representations from transformers (BERT)

BERT represents a significant advancement in NLP. It was introduced by Google researchers in 2018 and has set new standards in various NLP tasks due to its unique architecture. The main innovation of BERT is its bidirectional approach to language modeling. While other models, such as GPT, use unidirectional architectures that process text sequentially from left-to-right or right-to-left, BERT uses a bidirectional transformer architecture. This enables the model to predict masked words in a sentence by considering both the left and right context simultaneously, leading to a deeper understanding of language. The BERT model architecture is based on a multi-layer bidirectional transformer encoder. BERT has two sizes: BERT_BASE, with 12 layers, 768 hidden units per layer, and 12 self-attention heads, totaling 110 million parameters; and BERT_LARGE, with 24 layers, 1024 hidden units per layer, and 16 self-attention heads, totaling 340 million parameters. BERT's pre-training involves two main tasks: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP) [328]. In MLM, some tokens in a sentence are randomly masked, and the model is trained to predict these masked tokens based on the surrounding context. This helps the model understand the structure and meaning of the language. In NSP, the model is trained to understand the relationship between sentences by predicting if a sentence naturally follows another. This task is important for applications like question answering and natural language inference, where understanding sentence coherence is crucial. One of BERT's main advantages is its adaptability to various NLP tasks through fine-tuning. After pre-training, BERT can be fine-tuned with task-specific data by adding a simple output layer, requiring less data and computational resources than training a model from scratch. Fine-tuning has shown state-of-the-art performance on several benchmarks, including the GLUE benchmark, SQuAD, and SWAG. BERT's introduction has significantly impacted NLP, improving performance in tasks like sentiment analysis, named entity recognition, machine translation, and text summarization. BERT has also inspired further innovations, such as RoBERTa, ALBERT, and DistilBERT, building on its foundational concepts [329, 330] (Fig. 4).

3.2.4 A robustly optimized BERT pretraining approach (RoBERTa)

RoBERTa, an advanced version of the BERT model, was introduced by Facebook AI in 2019. This model builds on BERT's foundational architecture, aiming to enhance performance by refining the pretraining methodology and increasing the training scale. Its architecture is shown in Fig. 5. While RoBERTa retains the underlying transformer architecture of BERT,

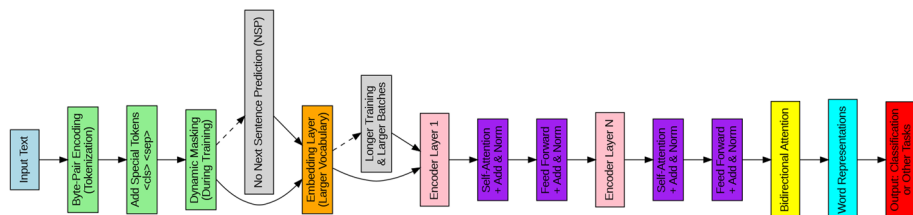


Fig. 5 Architecture of the RoBERTa model

it introduces several key modifications to the pretraining process. One primary change is the removal of the NSP objective. Researchers found that NSP did not significantly improve performance and could be omitted without degrading the model's effectiveness. By eliminating NSP, RoBERTa focuses solely on the MLM task, allowing the model to predict masked tokens more accurately. RoBERTa also extends the pretraining duration and increases the training data. RoBERTa is trained on ten times more data than BERT, using a dataset of 160GB compared to BERT's 16GB. This extended training corpus includes diverse sources, such as the Common Crawl news dataset, web text, and other large-scale corpora, ensuring comprehensive exposure to various linguistic patterns and contexts. Additionally, RoBERTa uses larger mini-batches and longer sequences during training. It uses a batch size of 8,000 sentences and trains for up to 500,000 steps, compared to BERT's 256-sentence batches and 1 million training steps. This increase in batch size and training duration enables RoBERTa to capture more nuanced relationships and dependencies within the data. RoBERTa also incorporates a more rigorous approach to hyperparameter optimization. The learning rate, batch size, and other critical parameters are finely tuned to ensure optimal performance. The model uses dynamic masking, where the masked tokens change during each epoch, providing a more robust learning experience. This contrasts with BERT's static masking, where the same tokens remain masked throughout the entire training process. Furthermore, RoBERTa leverages mixed-precision training techniques to improve computational efficiency and reduce memory usage. This approach allows the model to train on larger batches and longer sequences without compromising precision or performance, ultimately leading to a more powerful and efficient pretraining regimen. RoBERTa has been evaluated on various benchmarks and has outperformed BERT. On the GLUE benchmark, RoBERTa achieves higher scores on tasks such as sentiment analysis, text classification, and question answering. Additionally, RoBERTa excels in the SQuAD benchmark, setting new records for accuracy in both SQuAD 1.1 and SQuAD 2.0 [331].

3.2.5 XLNET

XLNET was introduced by researchers from Google AI Brain and Carnegie Mellon University in 2019. As a generalized autoregressive pre-training model, XLNET leverages the strengths of both autoregressive and auto-encoding models while addressing their limitations. BERT uses bidirectional context by reconstructing masked tokens but assumes independence among predicted tokens, which can lead to context fragmentation. XLNET addresses these limitations by integrating the autoregressive model's capacity for capturing dependencies and the auto-encoder's bidirectional context understanding. This integration is achieved through a permutation-based language modelling objective, where all possible permutations of the input sequence are considered during training. By doing so, XLNET effectively leverages

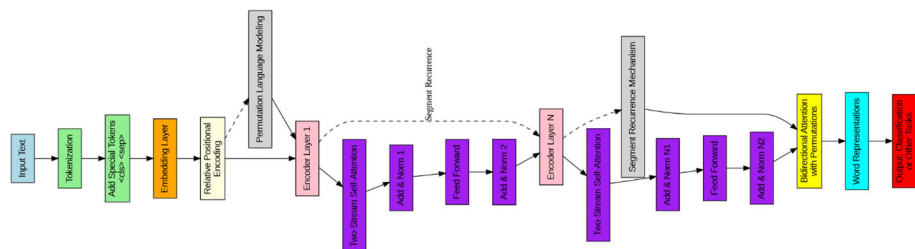


Fig. 6 Architecture of the XLNET model

bidirectional contexts while maintaining the dependency structure of autoregressive models. The core innovation of XLNET lies in its permutation-based training mechanism. Unlike BERT, which masks and predicts tokens in fixed positions, XLNET generates all possible permutations of the sequence and trains the model to predict tokens in these permuted contexts. This method ensures that each position can attend to tokens on both its left and right sides, capturing bidirectional context without sacrificing the autoregressive nature of the prediction task. This permutation language modelling objective enhances contextual representation and allows the model to learn dependencies across long-range sequences, improving its overall performance on various NLP tasks. The flexibility of permutation-based training enables XLNET to outperform traditional bidirectional models by avoiding the limitations of fixed masking strategies. XLNET further improves its contextual understanding through a segment-level recurrence mechanism inspired by Transformer-XL. This mechanism enables the model to capture dependencies beyond fixed-length segments by retaining contextual information across segment boundaries. By incorporating recurrence into the training process, XLNET can model long-term dependencies more effectively than conventional transformers, which are typically constrained by their fixed context window. XLNET's architecture is built upon the Transformer-XL framework, utilizing relative positional encoding and segment-level recurrence to enhance its ability to capture long-range dependencies. The model is pre-trained on a diverse corpus, including Wikipedia, BookCorpus, Giga5, ClueWeb 09, and Common Crawl, providing a broad range of linguistic contexts and improving generalization across tasks. During pre-training, XLNET employs dynamic data partitioning to handle variable sequence lengths efficiently. This approach ensures that the model can process sequences of different lengths without compromising computational efficiency. Moreover, the use of mixed-precision training techniques optimizes memory usage and speeds up the training process, enabling the handling of large-scale data [332]. Figure 6 shows the architecture of the XLNET model.

3.2.6 Efficiently learning an encoder that classifies token replacements accurately (ELECTRA)

ELECTRA was introduced by Google Research in 2020. ELECTRA replaces some tokens with plausible alternatives generated by a small generator model. The core innovation of ELECTRA is its discriminative objective, where the model learns to distinguish between original tokens and replaced tokens, rather than predicting the actual tokens. This approach is more efficient in terms of computational resources and sample utilization. The ELECTRA model consists of two components: the generator and the discriminator. The generator is a small MLM that predicts the probability distribution of possible replacements for masked

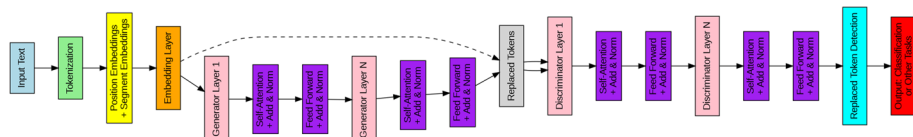


Fig. 7 Architecture of the ELECTRA model

tokens. The discriminator, which is the main focus of ELECTRA, is a transformer-based encoder trained to classify each token in the input sequence as either original or replaced. By leveraging a discriminative task, ELECTRA utilizes all input tokens rather than just the small subset of masked tokens used in MLMs, leading to more efficient learning. The replaced token detection task is the cornerstone of ELECTRA's pretraining strategy. During training, the generator replaces a certain percentage of tokens in the input sequence with alternatives sampled from its predicted distribution. The discriminator then takes this modified sequence and attempts to identify which tokens have been replaced. This binary classification task (original vs. replaced) is highly efficient and allows the model to leverage information from every token in the sequence. This pretraining method contrasts with the MLM objective, where only the masked tokens contribute to the loss computation. As a result, ELECTRA makes better use of the training data, leading to faster convergence and improved performance with the same or even fewer computational resources. ELECTRA has demonstrated superior efficiency compared to traditional MLM-based models. One of the significant advantages of ELECTRA is its reduced computational cost. Because the discriminator processes the entire input sequence without the need for masking and predicting each token, the pre-training phase is significantly faster and requires fewer resources. Furthermore, ELECTRA's pretraining objective leads to better parameter utilization, enabling smaller models to achieve performance levels comparable to or exceeding those of much larger models. For example, ELECTRA-Small achieves performance on par with BERT-Base, despite having only a fraction of the parameters. This efficiency makes ELECTRA particularly suitable for applications where computational resources are limited, such as edge devices or large-scale deployment scenarios. In various NLP benchmarks, ELECTRA has demonstrated strong performance, often surpassing BERT and other state-of-the-art models. On the GLUE benchmark, ELECTRA consistently achieves high scores across multiple tasks, including sentiment analysis, text classification, and natural language inference. Additionally, in the SQuAD benchmark for question answering, ELECTRA achieves competitive performance, highlighting its ability to understand and process complex queries and contexts [333]. Figure 7 shows the architecture of the ELECTRA model.

3.2.7 DistilBERT

DistilBERT is a lighter, faster variant of BERT developed by Hugging Face in 2019. It aims to reduce the computational cost and memory footprint of BERT while maintaining a high level of performance. DistilBERT achieves this through a process known as knowledge distillation, where a smaller "student" model is trained to replicate the behavior of a larger "teacher" model (in this case, BERT). This process involves training the student model to match the soft target probabilities produced by the teacher model, effectively transferring knowledge from the larger model to the smaller one. The primary advantage of DistilBERT is its reduced size and increased efficiency. DistilBERT has 40% fewer parameters than BERT, resulting in faster inference times and lower memory usage. Despite its smaller size, DistilBERT retains

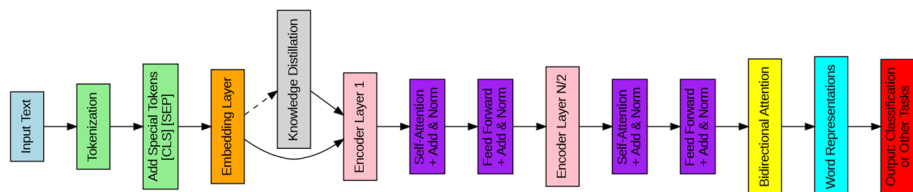


Fig. 8 Architecture of the DistilBERT model

97% of BERT's performance on various NLP benchmarks, making it an attractive option for applications where computational resources are limited. This efficiency is particularly beneficial for deploying NLP models on edge devices or in real-time applications where latency is critical. DistilBERT's architecture mirrors that of BERT but with fewer layers. While BERT-Base has 12 layers, DistilBERT has only 6. Despite this reduction, the model retains the self-attention mechanism and feedforward layers that are crucial for capturing contextual relationships in the input data. During training, DistilBERT uses a combination of supervised and unsupervised objectives. The supervised objective involves minimizing the cross-entropy loss between the student model's predictions and the ground-truth labels. The unsupervised objective involves minimizing the difference between the student and teacher models' predictions, effectively distilling the knowledge from the teacher to the student. This dual-objective training approach ensures that DistilBERT learns to generalize well from the available data while maintaining high fidelity to the teacher model's predictions. DistilBERT has been evaluated on various NLP benchmarks and has demonstrated strong performance across a range of tasks. On the GLUE benchmark, DistilBERT achieves competitive scores in tasks such as sentiment analysis, text classification, and natural language inference [334]. Figure 8 shows the architecture of the DistilBERT model.

3.2.8 A lite BERT (ALBERT)

ALBERT is an optimized version of BERT introduced by Google Research in 2019. It aims to address the limitations of BERT, such as its large memory footprint and slow training times, by introducing parameter reduction techniques that make the model more efficient without compromising performance. ALBERT achieves this through two main innovations: factorized embedding parameterization and cross-layer parameter sharing. Factorized embedding parameterization involves decomposing the large vocabulary embedding matrix into two smaller matrices. In BERT, the size of the embedding matrix is proportional to the vocabulary size multiplied by the hidden layer size, resulting in a large number of parameters. ALBERT reduces this size by first projecting the vocabulary embeddings into a lower-dimensional space and then projecting them back into the hidden layer space. This decomposition significantly reduces the number of parameters while maintaining the model's representational capacity. Cross-layer parameter sharing involves reusing the same parameters across multiple layers of the model. In BERT, each layer has its own set of parameters, leading to a large number of parameters overall. ALBERT reduces this redundancy by sharing parameters across layers, effectively reducing the model's parameter count without sacrificing depth or performance. This sharing can be applied to different components of the model, such as the attention parameters, feedforward parameters, or both. ALBERT also introduces a sentence-order prediction (SOP) task during pretraining, which helps the model better understand the coherence and ordering of sentences. This task involves predicting whether two consecutive sentences have

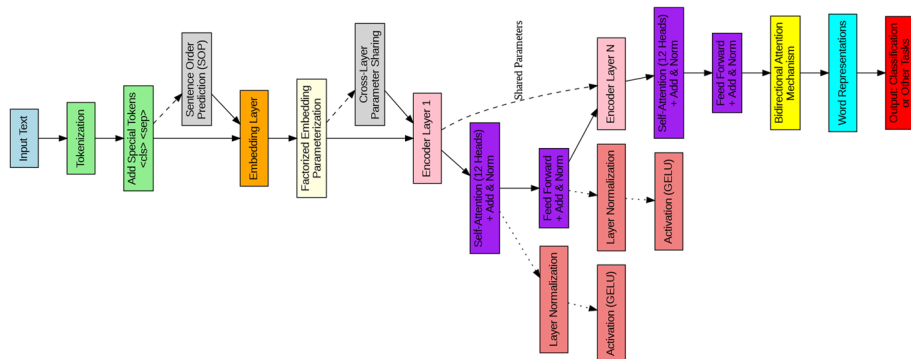


Fig. 9 Architecture of the ALBERT model

been swapped, providing a more challenging objective than the NSP task used in BERT. The SOP task helps the model capture inter-sentence relationships more effectively, leading to improved performance on downstream tasks. ALBERT has demonstrated significant improvements over BERT in terms of both efficiency and performance. The model achieves comparable or better results on various NLP benchmarks, including the GLUE benchmark and the SQuAD benchmark while using fewer parameters and less memory [335]. Figure 9 shows the architecture of the ALBERT model.

3.2.9 Text-to-text transfer transformer (T5)

T5 is a notable model introduced by Google Research in 2019. T5 frames every problem as a text-to-text task, so both the input and output are always text strings, no matter the specific task. For example, translation is handled as converting text from one language to another, and summarization as converting long text into shorter text. In sentiment analysis, T5 takes input like ‘sentiment analysis: I love this product!’ and generates an output like ‘positive,’ showing its ability to interpret and produce text-based sentiment labels. This uniform approach simplifies training and allows one model to be fine-tuned for various tasks without changing its structure. The text-to-text format helps T5 use transfer learning effectively, as the model is pre-trained on a wide range of tasks with a common input–output format. The T5 model uses an encoder-decoder setup where the encoder processes input text and the decoder generates output text. This design works well for the text-to-text framework, handling many sequence-to-sequence tasks. A key aspect of T5 is its use of self-supervised pretraining. The model is pre-trained on a large text corpus, including the Colossal Clean Crawled Corpus (C4), using a de-noising auto-encoder objective. In this method, parts of the input text are corrupted, and the model learns to reconstruct the original text. This pretraining allows T5 to develop a deep understanding of language, which can then be fine-tuned for specific tasks. The pretraining involves “span corruption,” where spans of tokens in the input text are replaced with a single mask token. The model must predict the missing spans based on the context [336]. Figure 10 shows the T5 model architecture.

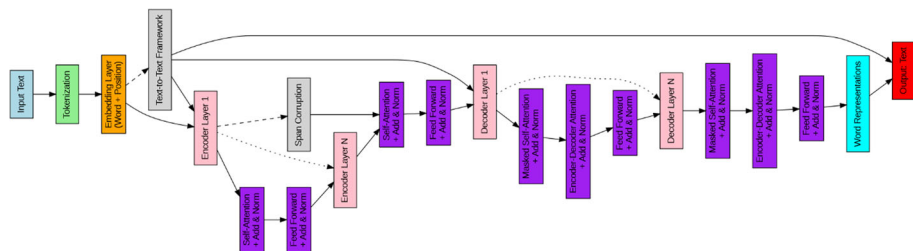


Fig. 10 Architecture of the T5 model

3.2.10 GPT

The GPT model marks a significant advancement in NLP. Introduced by OpenAI, the GPT series has set new standards in generating coherent and contextually relevant text. The original GPT was followed by GPT-2 and GPT-3, each improving the model's ability to understand and generate human-like text. GPT uses self-attention mechanisms to process input data, allowing the model to assess the importance of different words in a sequence and capture intricate dependencies within the text. The pretraining of GPT involves unsupervised learning on a large and diverse text corpus. During this phase, the model learns to predict the next word in a sentence based on the previous words within a context window. This autoregressive method ensures that the model captures the statistical properties of language and develops a robust understanding of syntax, semantics, and general world knowledge. GPT's architecture includes multiple layers of transformer blocks, each with a multi-head self-attention mechanism and position-wise feed-forward networks. GPT-2, for example, has 1.5 billion parameters spread across 48 layers. A key strength of GPT models is their ability to perform zero-shot, one-shot, and few-shot learning. In zero-shot learning, the model can generate plausible responses to prompts it has never seen during training. In one-shot and few-shot learning, GPT can use a few examples of a task to generate appropriate outputs. This adaptability is especially evident in GPT-3, which can handle tasks like translation, summarization, and question answering with minimal additional training [337]. Figure 11 shows the GPT-2 model architecture.

A detailed comparison of various transformer models, highlighting their features, strengths, and weaknesses, is provided in Table 6.

The diagram in Fig. 12 compares the mentioned models based on the number of param-

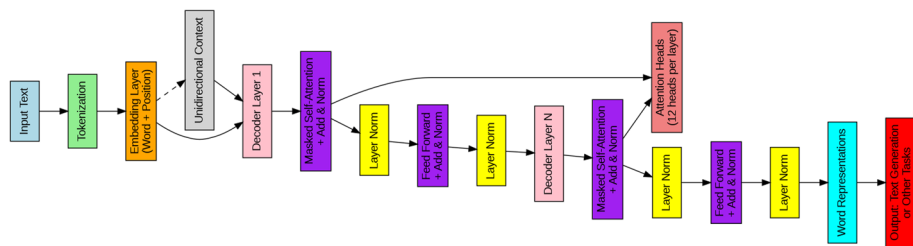


Fig. 11 Architecture of the GPT-2 model

Table 6 Comparison of various transformer models: Distinctive features, strengths, and weaknesses

Model	Description	Training objective	Strengths	Weaknesses	Unique features
BERT	It revolutionized NLP by pre-training deep bidirectional representations using unlabeled text	MLM and NSP	Superior at understanding context within text, making it excellent for tasks like question answering, named entity recognition, and sentiment analysis	Computationally intensive, particularly during inference; large model size requires substantial resources	Bidirectional context, NSP for understanding sentence relationships
RoBERTa	Enhances BERT by removing NSP and training with more data and longer sequences	MLM with dynamic masking, without NSP, trained on a larger corpus and for more steps	Outperforms BERT on several benchmarks due to extensive training, better handling of longer text sequences, and dynamic masking	Requires significant computational power for training, high resource consumption similar to BERT	Larger training data, dynamic masking, no NSP, longer training duration
XLNet	Generalized autoregressive pre-training that integrates ideas from Transformer-XL to model bidirectional context	PLM allowing context from all positions in a sequence	Combines the strengths of autoregressive and bidirectional models, effective for tasks requiring long-term dependencies and sequence understanding	More complex and longer training process, higher computational cost, and intricate architecture	Permutation-based LM, handles long-range dependencies
ELECTRA	Employs a discriminative training approach using a generator-discriminator setup	RTD, training the model to distinguish real tokens from plausible fakes generated by a small generator model	Highly efficient pre-training process, offers competitive performance with less computational effort, robust in resource-constrained environments	Less intuitive training objective, necessitates precise implementation of generator and discriminator components	RTD for efficient training, uses a small generator for dynamic negative sampling

Table 6 (continued)

Model	Description	Training objective	Strengths	Weaknesses	Unique features
DistilBERT	Distilled version of BERT, created to reduce the size while preserving much of BERT's original performance	Knowledge distillation from BERT, optimizing for size and speed	Significant reduction in model size and inference time, maintains approximately 97% of BERT's performance with half the number of parameters	Slight performance trade-off compared to full BERT, though substantially more efficient	60% smaller and faster than BERT, retains 97% of BERT's capabilities
ALBERT	Optimized to decrease memory footprint and increase training speed through innovative techniques	Factorized embedding parameterization and cross-layer parameter sharing, uses MLM and SOP	Drastically reduced parameters and memory usage, faster training, scalable to large datasets, maintains performance integrity	Complex parameter sharing and embedding techniques require careful tuning, potentially complex model architecture	Cross-layer parameter sharing, factorized embeddings, SOP for inter-sentence coherence
T5	Converts all NLP tasks into a text-to-text format, standardizing the approach across tasks	Seq2seq framework with MLM pre-train, handles tasks like translation, summarization, and question answering	Extremely versatile, excels in text generation and transformation tasks, state-of-the-art performance across various benchmarks	Large model sizes demand substantial computational resources, challenging to deploy and fine-tune in resource-limited settings	Unified text-to-text framework, highly versatile, state-of-the-art across diverse tasks
GPT	Emphasizes text generation through an autoregressive model, suitable for creative and generative tasks	Autoregressive language modeling, predicting the next token in a sequence given the previous tokens	Exceptional at text generation, including creative writing and dialogue systems, simple yet powerful training methodology	Limited to unidirectional context, less effective for tasks requiring deep understanding of context within entire sequences	Strong generative capabilities, large-scale models like GPT-3 demonstrate impressive zero-shot learning abilities

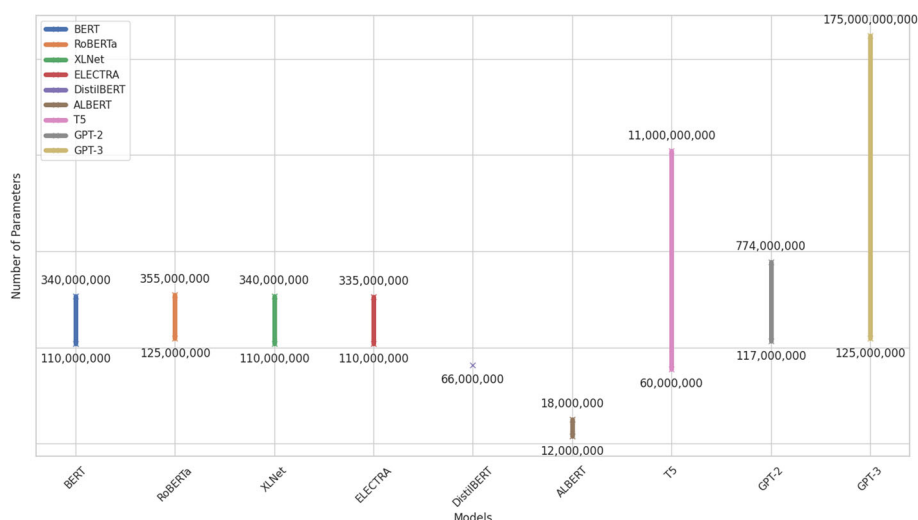


Fig. 12 Parameter count comparison of base and large models

eters. As shown, each model is classified into two types: base and large, each with different parameters. No information is available about the DistilBERT large model.

4 Experimental results

In this section, the datasets used for the results are introduced first. Then, the performance of well-known transformer models on each dataset is evaluated.

4.1 Datasets

Twenty-two well-known datasets from various domains, all focused on sentiment analysis, are utilized [338]. These datasets are detailed below:

- **Archeage [339]:** Comprises user reviews from the MMORPG Archeage, including positive and negative sentiments about gameplay, graphics, customer service, and updates. It is structured for sentiment analysis with labeled entries, making it valuable for understanding player satisfaction.
- **Debate [340]:** The Debate dataset consists of text data from online debate forums, curated for sentiment analysis. It includes labeled instances of positive, negative, and neutral sentiments, each representing a segment of a debate. This dataset is valuable for training and evaluating models in understanding sentiment within argumentative and opinionated contexts.
- **HCR [341]:** The HCR dataset is a sentiment analysis dataset curated specifically for the healthcare domain. It comprises a collection of patient reviews, feedback, and comments regarding their experiences with healthcare services, facilities, and medical professionals. The dataset covers various aspects of healthcare, including hospital visits, medical treatments, physician consultations, and overall patient satisfaction.

- Hobbit [339]: It consists of a diverse range of textual data extracted from various sources related to “The Hobbit” franchise, including book reviews, movie critiques, and discussions from online forums and social media platforms. The dataset covers sentiments expressed towards characters, plot elements, visual effects, and overall impressions of “The Hobbit” series.
- iPhone [339]: The iPhone dataset comprises a collection of user-generated content, predominantly consisting of reviews, comments, and sentiments expressed towards various iPhone models. The dataset includes a diverse range of sentiments, spanning from positive expressions of satisfaction with product features, performance, and user experience to negative sentiments regarding issues like product defects, customer service experiences, and pricing concerns.
- Irony [342]: The Irony dataset is specifically curated for sentiment analysis tasks, focusing on instances of irony in textual data. It comprises a collection of sentences or passages where the expressed sentiment contradicts the literal meaning or expectation, often used sarcastically or ironically.
- Movie [343]: The Movie dataset comprises a collection of user-generated reviews and ratings for various films. This dataset covers a wide range of movies across different genres and time periods, ensuring diversity in the sentiment expressed by users.
- Narr-KDML-2012 (Narr) [344]: It comprises a collection of narrative texts sourced from various domains, including literature, online forums, and social media platforms.
- Ntua [345]: These tweets reflect diverse sentiments expressed in everyday language. They cover a wide range of emotions, including joy, appreciation, disappointment, frustration, and longing.
- Person [343]: This dataset comprising a collection of social media posts that directly reference an individual. The primary objective of this dataset is to offer a detailed insight into the public’s emotional perception of the specified individual
- Sanders [346]: The dataset contains tweets related to four distinct topic keywords: “Apple,” “Google,” “Microsoft,” and “Twitter.” These tweets are sourced from the business domain.
- Sarcasm [342]: The Sarcasm dataset is curated specifically for sentiment analysis tasks, focusing on sarcastic comments.
- SemEval13 [347]: The SemEval-2013 Task 2 dataset is designed for sentiment analysis. It encompasses three subtasks: Message Polarity Classification, Aspect Category Detection, and Aspect Polarity Classification. In the Message Polarity Classification task, tweets are categorized as positive, negative, or neutral sentiments. Aspect Category Detection involves identifying the aspect or target category of sentiment within a tweet, like “food” or “service” in a restaurant review. Lastly, Aspect Polarity Classification focuses on determining the sentiment polarity (positive, negative, neutral) of each aspect category mentioned in the tweet.
- SemEval15 [348]: As part of the SemEval (Semantic Evaluation) competition, consists of annotated Twitter messages, covering a diverse range of topics and sentiments expressed in natural language.
- SemEval16 [349]: The SemEval-2016 dataset is a very large dataset. This dataset consists of tweets that express opinions and sentiments related to specific aspects of entities or products. This task involves identifying and analyzing opinions expressed towards particular aspects of entities or products mentioned in the text.
- SemEval17 [349]: The SemEval-2017 dataset for sentiment analysis is a widely used benchmark dataset in the field. Its domain is generic

- SemEval18 [350]: IT focuses on evaluating systems for sentiment analysis in the context of tweets. Released in 2018, it encompasses a diverse range of emotions and sentiments expressed in short, informal messages. Its domain is the Equity Evaluation Corpus.
- Sentiment140 [351]: The Sentiment140 dataset is a widely used resource for sentiment analysis tasks. The training portion of this dataset contains approximately 1.6 million tweets, making it one of the largest publicly available datasets for sentiment analysis. The tweets cover a diverse range of topics and are written in various styles, reflecting real-world language usage on social media platforms.
- SentiSternth [304]: The SentiStrength dataset comprises a collection of text samples annotated with sentiment strength scores, indicating the intensity of positive and negative sentiment expressed in each piece of text.
- STS-gold [352]: This dataset contains 2034 tweets and its domain is Generic.
- Target-dependent [353]: For the compilation of this dataset, specific keywords including “Elon Musk”, “Beyoncé”, “PlayStation”, “iOS”, and “Amazon” were utilized to retrieve tweets via the Twitter API. Following the acquisition of the tweets, human annotators manually assigned sentiment labels. The domain of this dataset is celebrities.
- VADER [354]: The VADER dataset consists of a comprehensive collection of text samples from various sources, including social media, news articles, and product reviews. Each text sample in the dataset is labeled with its corresponding sentiment score, indicating the degree of positivity, negativity, or neutrality conveyed by the text. Its domain is generic.

These datasets cover a wide range of sentiment analysis applications, including customer reviews, social media posts, movie reviews, and other contexts. This variety allows for a thorough assessment of the models’ performance across different sentiment analysis tasks. Figure 13 shows the distribution of positive and negative comments within these datasets, highlighting the balance and any inherent biases in the data.

4.2 Evaluation metrics

When the performance of machine learning algorithms is assessed, various evaluation metrics are often used. These metrics help understand the model’s performance and guide informed decision-making. Four essential evaluation metrics are explored in this article: Accuracy, Precision, Recall, and F1-score [8, 355].

- Accuracy measures the overall correctness of predictions. It answers the question: “What proportion of all predictions were correct?” Accuracy is commonly used when the class distribution is balanced but can be misleading in imbalanced datasets, where high accuracy can be achieved even if all negatives are predicted. It ignores the impact of false positives and false negatives. Mathematically, it is defined as Eq. 1:

$$Accuracy = \frac{TruePositive + TrueNegative}{count\ of\ samples} \quad (1)$$

- Precision focuses on the accuracy of positive predictions. It answers: “Of all predicted positives, how many were truly positive?” Precision is crucial when avoiding false positives is essential. High precision may lead to low recall (missing true positives). It is sensitive to false positives. The formula is Eq. 2:

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive} \quad (2)$$

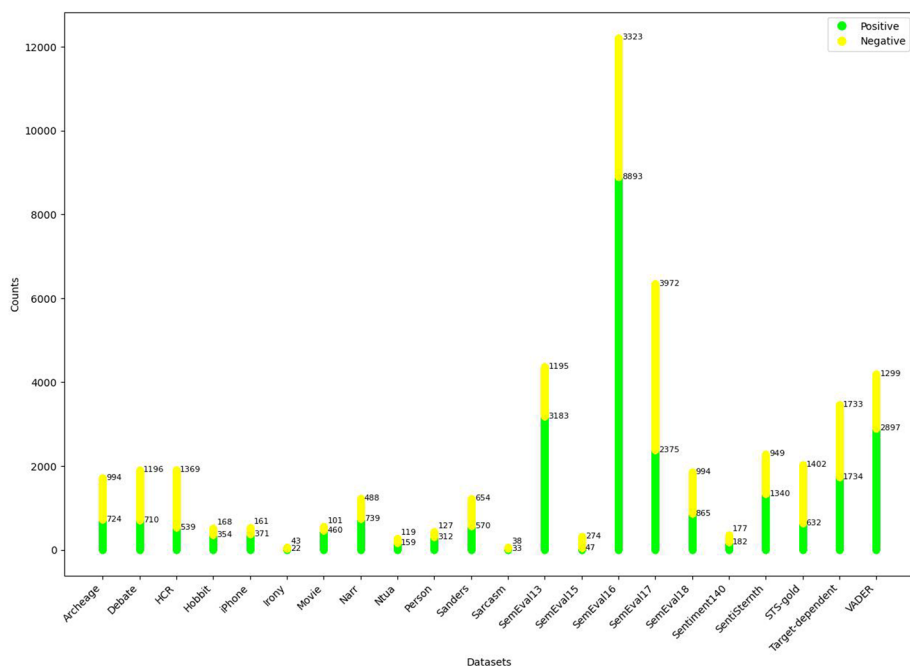


Fig. 13 Distribution of negative and positive samples in the dataset

- Recall, also known as sensitivity, assesses the ability to capture actual positives. It answers: “Of all true positives, how many were correctly identified?” High recall may lead to low precision (more false positives) and is sensitive to false negatives. The formula is in Eq. 3:

$$\text{Recall} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalseNegative}} \quad (3)$$

- The F1-score balances precision and recall. It is the harmonic mean of the two and is useful when class distributions are uneven. It combines both precision and recall, providing a single metric to evaluate model performance. Its formula is in Eq. 4.

$$F1 = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

4.3 Results

In this section, the performance of various transformer models, including BERT, RoBERTa, XLNet, ELECTRA, DistilBERT, ALBERT, T5, and GPT, across different datasets, is compared and evaluated. Each model has been fine-tuned on specific datasets to optimize performance for sentiment analysis tasks. For example, BERT, RoBERTa, XLNet, ELECTRA, DistilBERT, and ALBERT have been fine-tuned on the Sentiment140 dataset, a well-known benchmark for sentiment analysis. In contrast, the GPT model has been fine-tuned using Amazon website comments, and the T5 model using StanfordNLP dataset comments. Table 7 presents the hyperparameters used for tuning each model. All models, except XLNET, GPT2, and T5, were trained with a learning rate of 1e-5, a batch size of 16, a seed value of

Table 7 Hyperparameters used for tuning each model

	Learning rate	Training batch size	Seed	Optimizer	Epochs
BERT	1e-05	16	42	Adam betas = (0.9,0.999), epsilon = 1e-08	5
RoBERTa	1e-05	16	42	Adam betas = (0.9,0.999), epsilon = 1e-08	5
DistilBERT	1e-05	16	42	Adam betas = (0.9,0.999), epsilon = 1e-08	5
XLNET	3e-05	16	42	Adam betas = (0.9,0.999), epsilon = 1e-08	1
ALBERT	1e-05	16	42	Adam betas = (0.9,0.999), epsilon = 1e-08	5
ELECTRA	1e-05	16	42	Adam betas = (0.9,0.999), epsilon = 1e-08	5
GPT2	2e-05	16	42	Adam betas = (0.9,0.999), epsilon = 1e-08	2
T5	1e-05	16	42	Adam betas = (0.9,0.999), epsilon = 1e-08	10

42, and the Adam optimizer with betas (0.9, 0.999) and epsilon of 1e-8. XLNET used a higher learning rate of 3e-5 and was trained for 1 epoch. GPT2 was trained with a learning rate of 2e-5 over 2 epochs, while T5 had a longer training period of 10 epochs with the standard learning rate. These variations in hyperparameters reflect the optimization needed for each model to achieve the best performance.

Table 8 displays the F1 values obtained on 22 datasets using eight models. The highest F1 values are shown in bold.

As shown in Table 8, the T5 model consistently ranks highest in multiple datasets, including Archeage, Debate, HCR, Hobbit, Movie, Narr, SemEval17, SemEval18, and Senti-ment140. This suggests T5's versatility and ability to generalize well across different domains. The XLNet model shows outstanding performance in the iPhone, Irony, and SentiStrength datasets, indicating its effectiveness in understanding irony and product-related sentiments. ELECTRA excels in the Sanders and SemEval16 datasets. RoBERTa leads in the Ntua and STS-gold datasets, highlighting its strength in general sentiment analysis.

BERT and DistilBERT frequently rank lowest in several datasets. This pattern indicates that while they are computationally efficient, they may lack the depth of understanding required for more complex sentiment analysis tasks. The SemEval15 dataset shows DistilBERT as the top performer with an F1 score of 42.47, while ALBERT scores the lowest at 32.17, suggesting that simpler models can outperform more advanced ones in specific

Table 8 F1 scores of eight models across 22 datasets

	BERT	RoBERTa	DistilBERT	XLNET	ALBERT	ELECTRA	GPT2	T5
Archeage	73.09	74.72	70.10	75.80	71.28	76.65	76.66	79.85
Debate	63.59	69.02	65.59	65.85	67.73	70.17	68.36	74.71
HCR	49.58	54.25	49.69	50.64	51.53	56.25	56.17	59.24
Hobbit	76.86	77.40	78.45	79.71	74.93	77.79	80.30	84.93
iPhone	77.51	75.24	77.86	84.27	77.33	81.82	81.39	79.88
Irony	69.09	69.57	62.96	75.47	65.52	73.47	72.00	73.17
Movie	88.43	88.14	87.88	89.22	88.25	89.95	89.51	93.36
Narr	88.56	91.78	90.29	91.26	89.80	93.14	89.99	93.81
Ntua	86.54	89.04	86.62	88.75	88.96	88.68	78.98	87.21
Person	85.53	85.57	85.80	86.26	86.33	87.54	89.80	87.97
Sanders	75.11	83.38	78.56	80.89	79.00	84.30	80.03	81.60
Sarcasm	73.81	72.50	74.07	73.81	72.29	73.17	73.42	73.97
SemEval13	87.40	87.89	87.09	90.03	86.89	89.99	89.17	89.76
SemEval15	38.71	41.56	42.47	34.82	32.17	38.97	33.33	38.92
SemEval16	84.79	85.61	84.12	87.62	83.77	88.16	87.98	86.59
SemEval17	69.25	75.79	71.22	72.36	69.20	76.56	79.23	83.41
SemEval18	81.49	85.85	81.97	83.99	78.69	85.71	82.20	86.92
Sentiment140	86.26	89.01	83.78	87.63	81.74	87.60	85.01	89.27
SentiSternth	82.43	85.63	83.28	86.44	82.57	85.57	83.65	83.70
STS-gold	87.38	91.38	87.32	89.32	88.69	89.97	74.44	81.63
Target-dependent	73.67	75.42	74.67	76.77	73.07	75.35	81.20	80.39
VADER	85.10	83.47	84.29	83.91	86.85	83.75	83.85	83.52

contexts. In the HCR dataset, T5's leading performance (59.24) and BERT's lagging performance (49.58) reflect the need for advanced contextual understanding in healthcare-related sentiments.

Figure 14 provides a comparative analysis of the F1 rankings of various transformer models across 22 datasets. The y-axis represents the rank of each model, with a lower rank indicating better performance. BERT shows consistent performance across the datasets with a median rank of around 6. However, its range extends from 2 to 8, indicating that while it performs reasonably well overall, it struggles to consistently achieve top ranks. RoBERTa's performance is quite variable, with a median rank of 4, suggesting that it frequently performs better than BERT. However, its wide interquartile range from 2 to 6 indicates significant variability, suggesting RoBERTa's performance depends on the dataset's characteristics. DistilBERT has a median rank of 5, with a relatively narrow interquartile range from 5 to 7, indicating more consistent performance compared to RoBERTa, though it does not often achieve top ranks.

XLNet's performance shows a relatively high median rank around 4. The interquartile range from 3 to 4 indicates strong performance, often achieving high ranks across datasets. ALBERT has a median rank of 7 but shows wide variability, with ranks ranging from 5 to 8. This suggests that while ALBERT can achieve top ranks, its performance is less predictable and highly dependent on the dataset. ELECTRA has a median rank of 3 with a narrow interquartile range from 2 to 4, indicating consistent performance across most datasets. GPT-2 shows a wide range of performance with a median rank of 4. Its interquartile range from 2 to 5 indicates variability, suggesting that while GPT-2 can perform well, its effectiveness varies significantly across different datasets. Finally, T5 has the lowest median rank of 2,

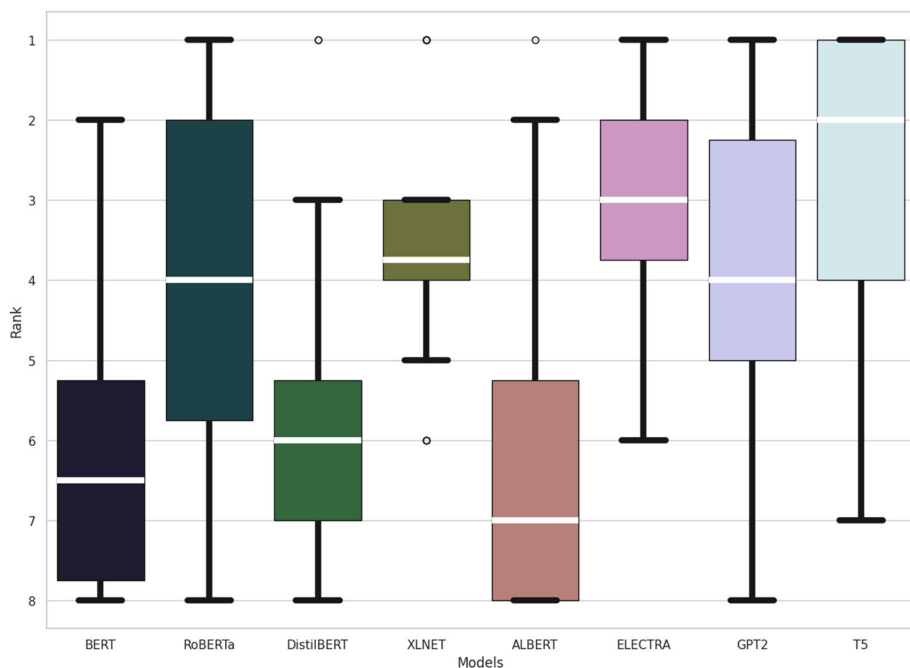


Fig. 14 Comparative analysis of F1 rankings for transformer models across 22 datasets

with an interquartile range from 1 to 4, indicating that T5 consistently achieves top ranks across most datasets.

Figure 14 also highlights the presence of outliers in the performance rankings of several models, depicted with circles. These outliers represent instances where a model's performance deviates significantly from its typical range. For example, XLNet has outliers indicating instances where it ranked notably lower or higher than usual, reflecting occasional poor or top performance on certain datasets. Table 9 shows the accuracy values for different models. In cases where the data distribution is uneven or there are unbalanced classes, the model may perform well in accuracy but have a lower F1 because the ability to correctly distinguish positive and negative samples (sensitivity and specificity) is lower. For example, in datasets such as SemEval15 and HCR, the models have high accuracies but significantly lower F1 values. This difference shows that the models are less balanced in correctly identifying positive and negative samples and achieve high accuracy only based on the correct prediction of most samples. Therefore, F1 values provide a more comprehensive assessment of model performance in the face of unbalanced data and show how well a model can perform under different conditions.

Table 10 compares the recall of various sentiment analysis models across different datasets. Recall measures a model's ability to correctly identify positive instances and varies significantly among the models and datasets. Overall, models like XLNET and ELECTRA tend to perform well across many datasets, likely due to their architectures that effectively capture context. XLNET, for example, often outperforms others, possibly because of its ability to capture long-term dependencies, which is crucial for understanding complex sentences. However, no model consistently outperforms across all datasets. RoBERTa and DistilBERT models did not achieve the best recall for any dataset, indicating their limitations in finding positive examples. In datasets containing sarcasm, XLNET and BERT models perform better than others, as they are more capable of recognizing sarcastic content with a positive meaning. Despite its generally strong performance, T5 only provides the best results in two specific datasets according to the recall criterion, highlighting its weakness in detecting positive samples across a broader range of data. Overall, ELECTRA has achieved the best performance in terms of recall, demonstrating its effectiveness in identifying positive instances.

Figure 15 illustrates the values of true positive (TP), true negative (TN), false positive (FP), and false negative (FN) for different models. In the TP and TN charts, the model achieving the highest value is highlighted, with the corresponding number labeled. Conversely, in the FP and FN charts, the model with the lowest value is marked, along with the respective number. The analysis reveals that the T5 model has attained the highest TP value across 16 out of 22 datasets, indicating superior performance in identifying positive samples. On the other hand, the ELECTRA model has achieved the highest TN value on 8 datasets, while the T5 model has managed this only on 2 datasets. Furthermore, ELECTRA has the lowest FP value on 8 datasets, signifying its efficiency in minimizing false positives and accurately predicting true negatives, thereby excelling in identifying negative samples. Additionally, the FN chart indicates that the T5 model has the lowest FN value on 16 datasets. This suggests that while T5 excels in predicting positive samples, it underperforms in predicting negative samples, as it tends to label negative instances less frequently. This duality implies that T5 is more likely to miss negative samples despite its high accuracy in predicting positive ones.

Table 9 Accuracy values of eight models across 22 datasets

	BERT	RoBERTa	DistilBERT	XLNET	ALBERT	ELECTRA	GPT2	T5
Archeage	74.97	80.15	72.29	78.41	73.46	79.22	80.15	84.11
Debate	70.20	78.33	67.58	70.67	72.35	73.45	74.45	81.85
HCR	61.95	72.06	66.14	59.85	60.17	67.71	70.96	76.99
Hobbit	69.54	70.69	71.26	72.99	67.05	69.92	75.10	80.08
iPhone	71.43	70.68	71.99	78.95	70.68	76.69	75.75	75.56
Irony	73.85	78.46	69.23	80.00	69.23	80.00	78.46	83.08
Movie	82.17	81.82	81.46	83.24	81.82	84.31	83.96	89.48
Narr	86.31	90.38	88.43	89.41	87.86	91.69	88.26	92.75
Ntua	84.89	88.13	84.89	87.05	87.41	87.05	76.26	85.97
Person	79.04	79.95	79.50	80.18	79.73	82.23	85.19	83.37
Sanders	76.23	85.21	79.58	81.05	79.33	84.97	81.29	84.23
Sarcasm	69.01	69.01	70.42	69.01	67.61	69.01	70.42	73.24
SemEval13	82.48	83.53	82.25	85.95	81.73	85.82	84.58	85.88
SemEval15	70.40	71.96	73.83	54.52	51.40	62.93	58.88	64.8
SemEval16	78.65	80.62	78.12	82.61	77.36	83.38	83.00	82.02
SemEval17	71.70	80.54	74.62	74.60	71.20	79.74	82.59	87.05
SemEval18	81.76	86.82	82.63	84.19	78.64	85.58	82.73	87.47
Sentiment140	86.07	88.86	83.29	87.19	81.34	87.47	84.68	89.42
SentiSternth	78.64	83.40	79.90	83.53	79.03	82.26	81.13	81.91
STS-gold	91.69	94.64	91.54	92.97	92.67	93.41	81.02	87.56
Target-dependent	71.24	74.68	72.19	75.02	70.35	73.52	80.21	80.44
VADER	80.65	79.36	79.98	79.53	82.44	79.29	80.03	79.91

The best value obtained on each dataset is shown in bold

Table 10 Recall values of eight models across 22 datasets

	BERT	RoBERTa	DistilBERT	XLNET	ALBERT	ELECTRA	GPT2	T5
Archeage	80.66	69.61	77.07	80.25	78.18	80.94	77.35	74.72
Debate	69.86	64.79	82.96	75.92	77.89	83.8	74.08	71.97
HCR	66.23	58.63	59.18	72.91	74.95	73.47	65.86	59.18
Hobbit	74.58	74.01	77.12	78.25	72.6	77.68	74.86	82.77
iPhone	70.62	63.88	70.62	80.86	71.7	75.2	76.01	69.54
Irony	86.36	72.73	77.27	90.91	86.36	81.82	81.82	68.18
Movie	83.04	82.39	81.96	84.57	83.26	85.65	83.48	90.22
Narr	87.96	89.17	89.31	91.88	88.77	93.64	87.55	91.2
Ntua	84.91	84.28	85.53	89.31	88.68	88.68	77.99	83.65
Person	87.18	83.65	87.18	87.5	90.06	87.82	91.67	85.58
Sanders	77.02	79.65	80.35	86.14	83.51	86.67	80.53	75.09
Sarcasm	93.94	87.88	90.91	93.94	90.91	90.91	87.88	81.82
SemEval13	83.6	82.19	82.31	87.21	83.32	87.65	87.28	85.11
SemEval15	63.83	68.09	65.96	82.98	78.72	80.85	70.21	76.6
SemEval16	81.76	79.2	79.61	84.5	80.28	85	85.51	79.7
SemEval17	85.14	81.39	83.92	88.84	86.48	88.42	88.72	86.99
SemEval18	86.24	85.9	84.86	89.13	84.74	92.95	85.66	89.48
Sentiment140	86.26	89.01	85.16	89.56	82.42	87.36	85.71	86.81
SentiSternth	85.6	84.48	85.52	89.7	84.85	89.85	82.46	79.33
STS-gold	92.56	91.46	93.67	94.62	92.41	95.09	88.92	88.92
Target-dependent	80.45	77.68	81.95	82.53	80.45	80.91	85.41	80.16
VADER	80.05	75.46	77.77	77.32	83.98	77.29	75.08	73.73

The best value obtained on each dataset is shown in bold

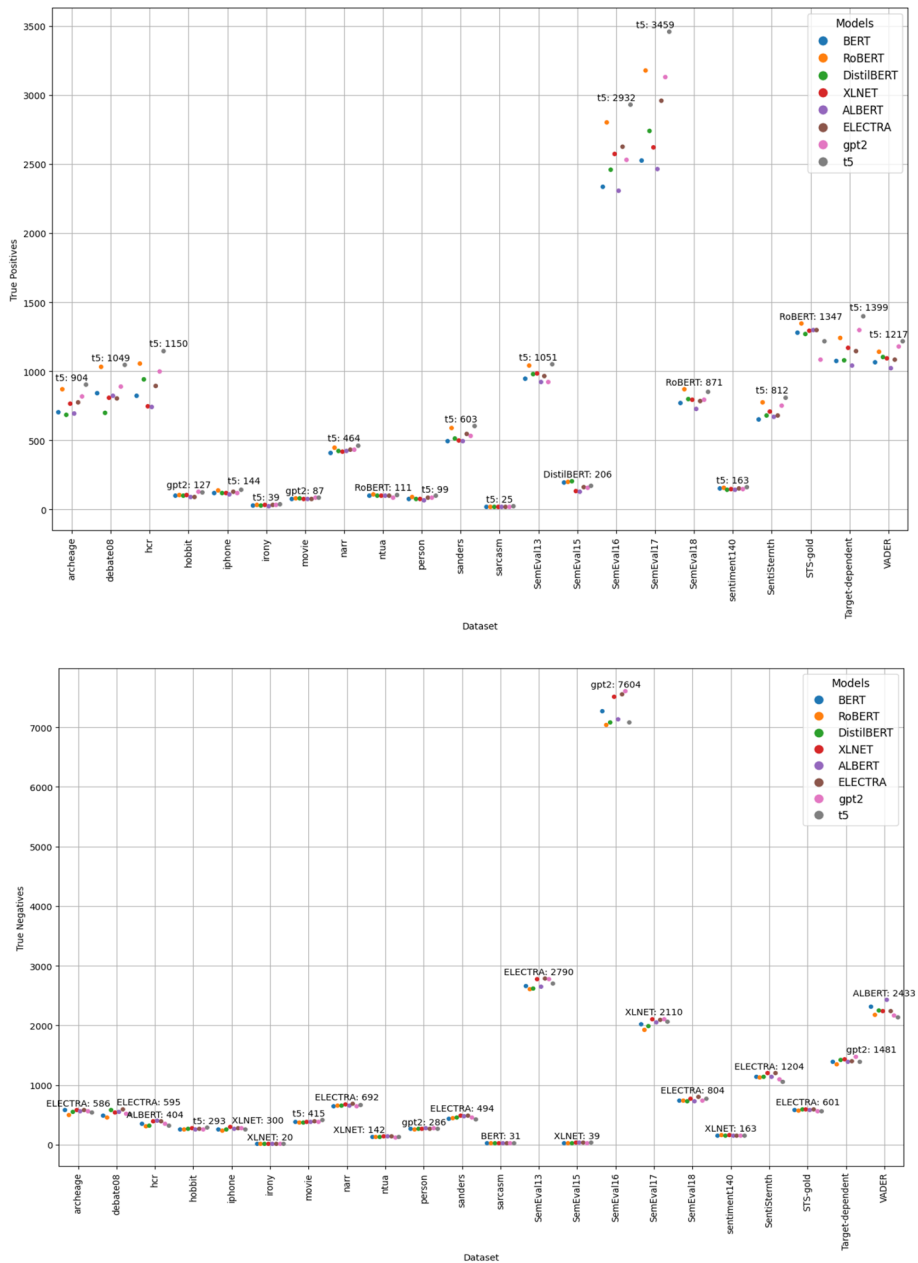


Fig. 15 Distribution of true positive (a), True negative (b), False positive (c), and False negative (d) instances across different models

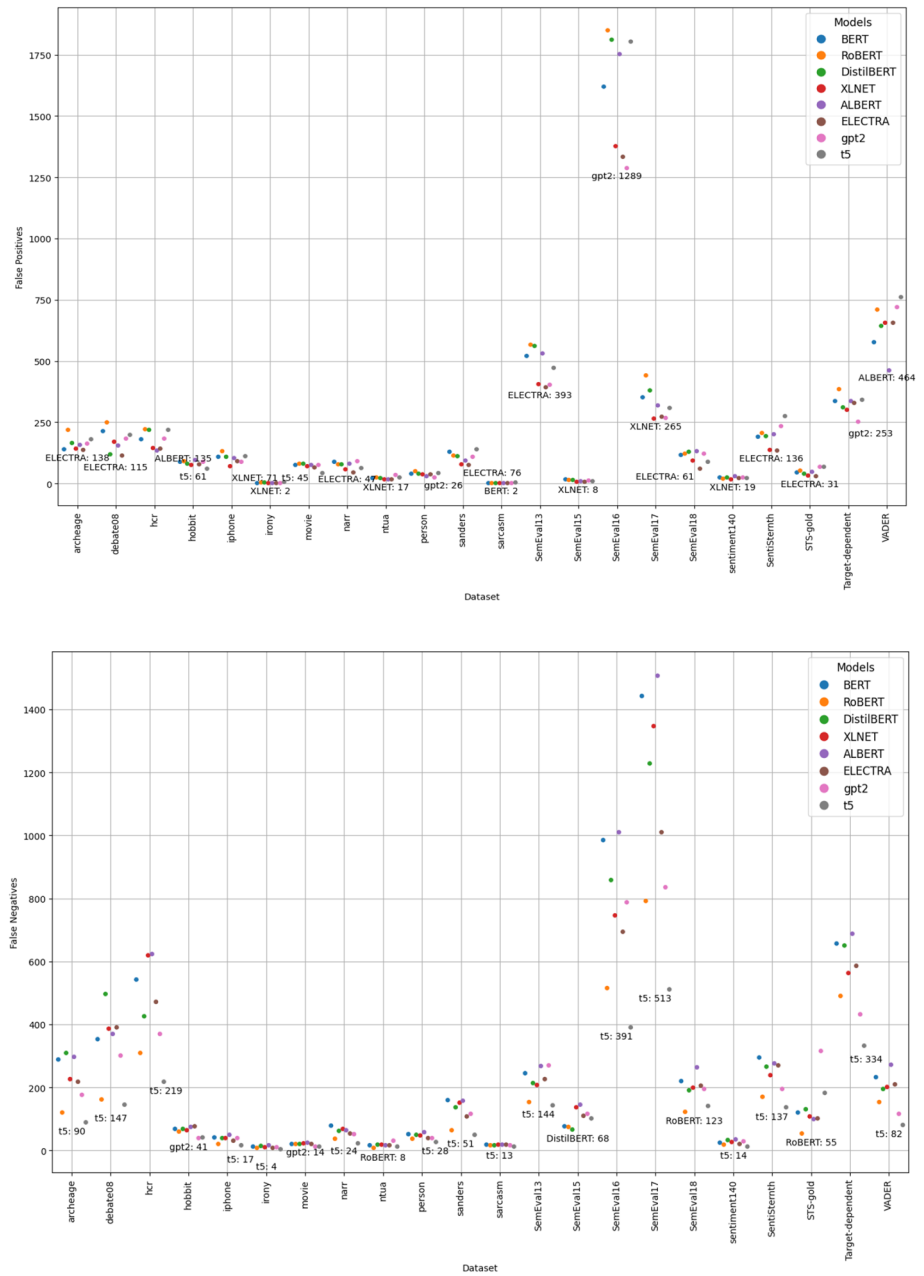


Fig. 15 continued

5 Limitations and future works

Transformer models, while powerful, have several limitations. They are computationally intensive and require significant resources for both training and inference, making them less accessible for organizations with limited resources. The self-attention mechanism, which is central to their functionality, leads to high memory usage, especially for long sequences. Additionally, these models need large amounts of training data to perform well, which can be challenging to acquire and preprocess. Transformers often act as “black boxes,” with limited interpretability, making it difficult to understand how they arrive at certain predictions. This lack of transparency can hinder trust and adoption in critical applications. Moreover, transformer models can inherit and propagate biases present in their training data, raising concerns about fairness and ethical use. Finally, their performance can degrade when applied to domains different from the ones they were trained on, necessitating domain-specific fine-tuning.

Future work in the field of transformers and sentiment analysis should focus on several key areas. Developing more computationally efficient architectures that require less power and memory without compromising performance is crucial. Enhancements in handling long sequences efficiently, through methods such as sparse attention or hierarchical models, are needed. To reduce the dependence on large datasets, exploring techniques like few-shot learning, active learning, and synthetic data generation will be beneficial. Addressing biases through diverse training datasets, fairness constraints, and bias correction techniques should be prioritized. Enhancing the ability of models to generalize across different domains through domain adaptation, multi-task learning, and transfer learning can broaden their applicability. Developing multilingual and cross-cultural models by training on diverse datasets that represent various linguistic and cultural nuances is also important. Lastly, integrating human feedback into the training loop to enhance human-AI collaboration can improve the accuracy and reliability of these models. Additionally, exploring specific combinations of models like ELECTRA and T5, given their different strengths in handling positive and negative examples, could lead to more robust and versatile sentiment analysis systems.

6 Conclusion

Sentiment analysis, a key term in natural language processing, has gained significant attention due to its complex applications in various fields. This study provides a comprehensive review of sentiment analysis, covering its definition, challenges, and applications, as well as discussing different methods used. Special developments in transformers and transfer learning are also reviewed. The experimental results showed that the T5 model consistently outperforms others across multiple datasets, demonstrating its versatility and strong generalization capabilities. This suggests that T5 is particularly well-suited for various sentiment analysis tasks, making it a robust choice for real-world applications. Conversely, models like XLNet performed exceptionally well in specific contexts, such as detecting irony and analyzing product-related sentiments, highlighting the importance of choosing models based on the specific characteristics of the dataset. While BERT and DistilBERT offer computational efficiency, they often ranked lower in performance. ELECTRA's high performance in reducing false positives and accurately predicting true negatives highlights its effectiveness in identifying negative sentiments, which is crucial for applications requiring high precision.

Declaration of generative AI and AI-assisted technologies in the writing process:

During the preparation of this work the authors used ChatGPT, Quillbot, and Grammarly in order to rephrase and edit the text grammatically and in terms of typo. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declarations

Conflict of interest The authors declare no competing interests.

Ethics approval Not applicable.

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